IES

GenAI for Engineers

Boosting productivity and engineering data analysis with Microsoft 365 Copilot

1. **Provider · IES**
2. Course · One day · 09:00–17:00
3. Approach · Understand → prompt → produce → verify



Learning outcomes · LO1–LO4

Methods, documents, prompts and agents

1 **LO1** Understand FMEA, Six Sigma DMAIC and SPC before applying them.

2 **LO2** Automate engineering documents with Microsoft 365 Copilot.

3 **LO3** Apply R.I.S.E. and E-MAP with iterative prompting.

4 **LO4** Build and deploy a multi-step Copilot Agent.

Learning outcomes · LO5–LO7

Reasoning agents, SharePoint grounding and secure use

5

LO5 Use Researcher and Analyst for research and data analysis.

6

LO6 Integrate SharePoint as a cited engineering knowledge base.

7

LO7 Apply the complete 7-Step AI Security Framework.

Today’s route

Six modules follow the supplied lesson plan

1 **09:00**
M1 · Copilot ecosystem

2 **10:00**
M2 · Prompting

3 **11:15**
Tea break

4 **11:30**
M3 · FMEA → DMAIC → SPC

5 **13:00**
Lunch

6 **14:00**
M4 · Agents

7 **15:15**
Tea break

8 **15:30**
M5 · SharePoint

9 **16:15**
M6 · Security

10 **16:45**
Summary and close

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MODULE 1

The Copilot Ecosystem for Engineers

Choose the right Copilot surface for the engineering job



Module 1 learning outcomes

By the end of this module, participants can:

1	**Explain**	Distinguish GenAI, Copilot and an agent.
2	**Choose**	Match Chat, Word, Excel, PowerPoint and SharePoint to a task.
3	**Apply**	Turn a messy pump problem into a clear brief.
4	**Verify**	Keep evidence and engineering decisions visible.

AI, GenAI and Copilot are related

Start with three simple definitions

AI

Software performs tasks that usually need human judgement.

Generative AI

Creates a new draft from instructions and context.

Microsoft 365 Copilot

Brings GenAI into work apps and organisational data.

Where productivity improves

Copilot shortens repetitive preparation and first-draft work

1 **Find**

Locate relevant content

2 **Structure**

Turn notes into fields

3 **Calculate**

Suggest formulas and summaries

4 **Draft**

Create a report or slide

5 **Review**

Check evidence and approve

The Microsoft 365 Copilot ecosystem

Different surfaces support different engineering outputs

1 Copilot Chat

Questions and briefs

2 Word

Specifications and reports

3 Excel

Tables, formulas and charts

4 PowerPoint

Management summaries

5 Agents

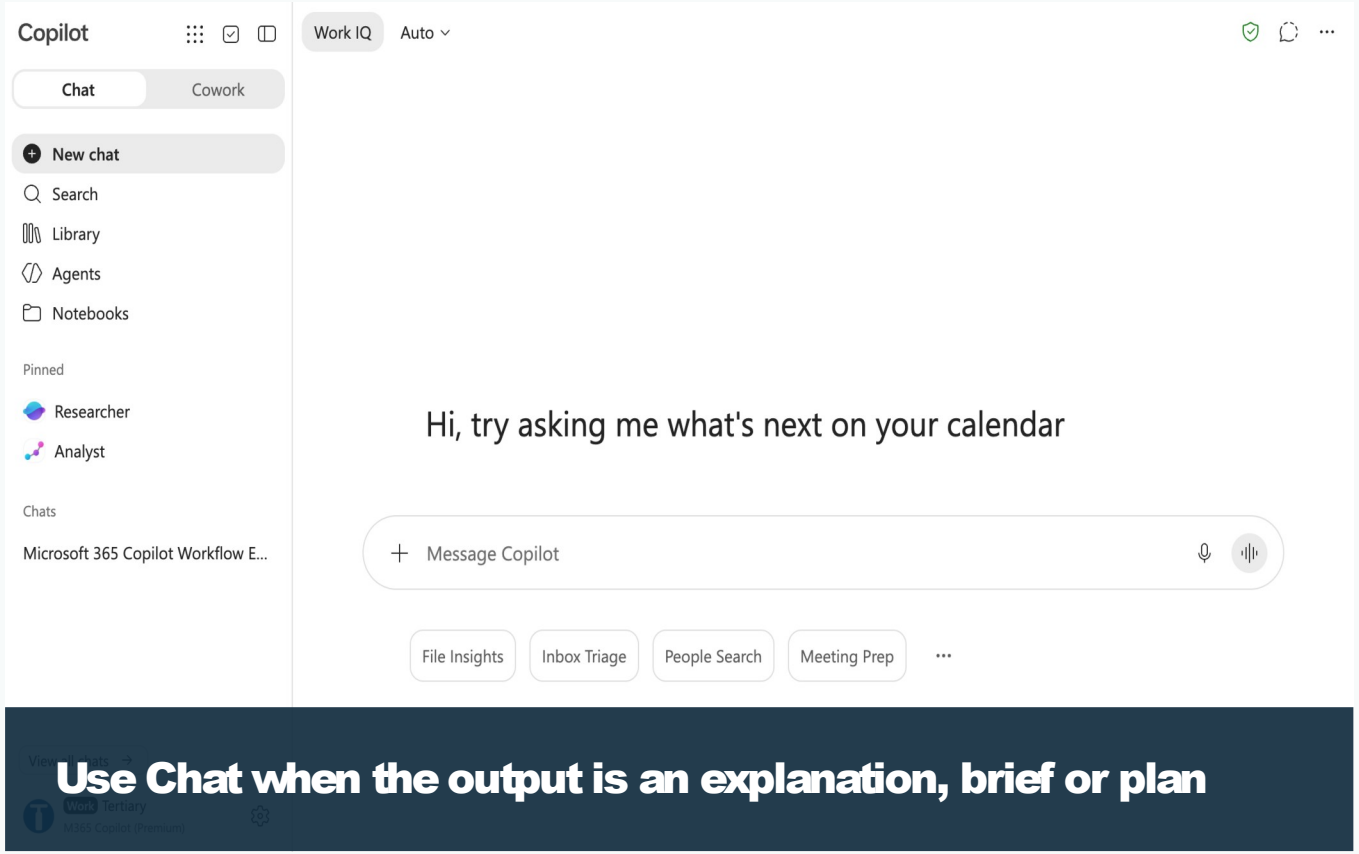
Repeatable workflows

6 SharePoint

Grounded knowledge

Copilot Chat · shape the question

Use Chat when the output is an explanation, brief or plan



1 Input

Messy fault notes and known facts

2 Copilot helps

Organise facts, assumptions and open questions

3 Engineer checks

No invented values or conclusions

Copilot in Word · draft structured documents

Move from source notes to an editable engineering document

1

Source

Requirements or inspection notes

2

Prompt

Sections, tone, limits and evidence

3

Draft

Spec, report, SOP or method statement

4

Review

Clauses, numbers and approvals

Copilot in Excel · analyse structured data

Use named tables, explicit formulas and independent checks

1

Prepare

Headers, units and table name

2

Ask

Calculation and chart instructions

3

Inspect

Formulas, ranges and missing values

4

Verify

One result independently

Copilot in PowerPoint · communicate decisions

Turn checked findings into a short audience-specific story

1

Message

One decision or question

2

Evidence

Two or three supporting facts

3

Visual

Chart, process or comparison

4

Boundary

What remains unknown

Copilot in Teams · turn discussion into action

Convert meeting content into follow-up work

1

Meeting

Transcript or notes

2

Draft

Decisions and actions

3

Assign

Owner and due date

4

Review

Confirm wording with participants

SharePoint · supply governed context

A knowledge base helps Copilot answer from controlled sources

1

Documents

Standards, manuals, reports

2

Permissions

Who may see each source

3

Agent

Searches the selected scope

4

Answer

Returns citations to open and check

Agents · reuse a multi-step job

An agent combines instructions, knowledge and a repeatable workflow



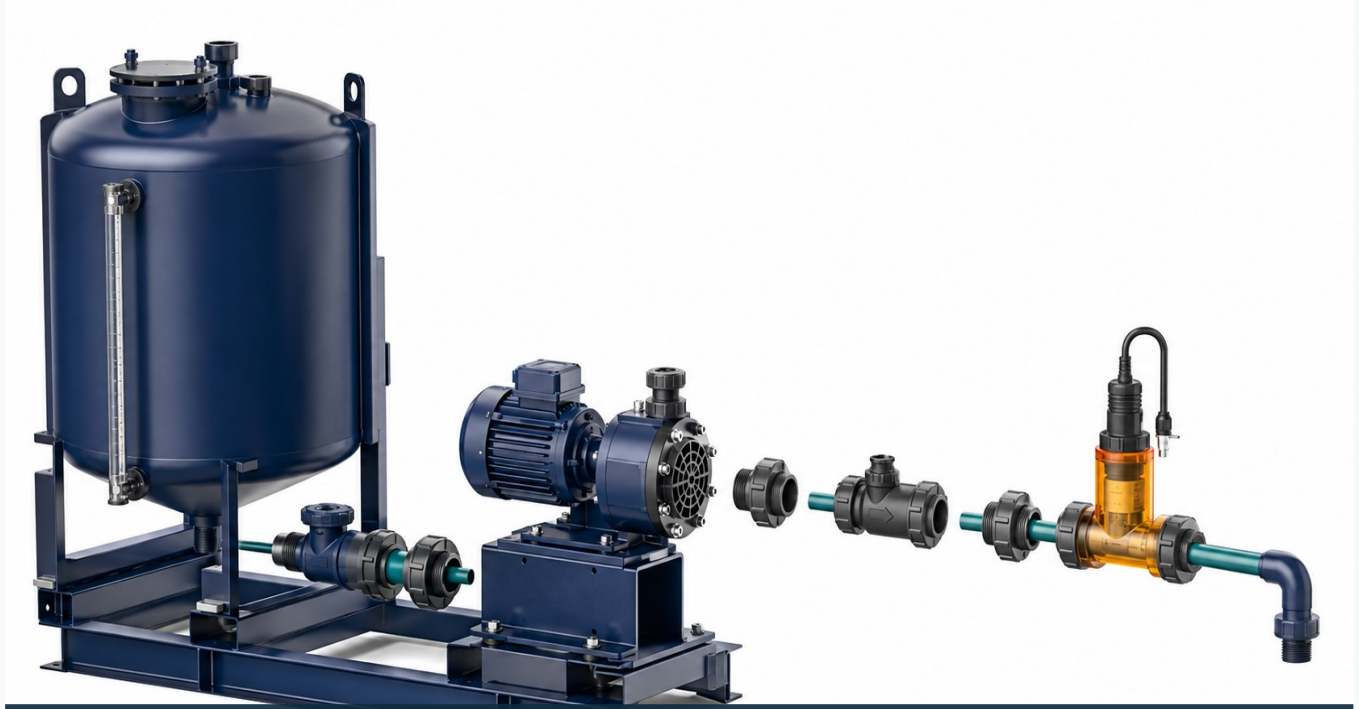
Choose the surface from the output

Start with what the engineer needs to produce

Question or brief	Editable report	Calculation or chart	Decision story	Repeatable workflow	Cited internal answer
Copilot Chat	Word	Excel	PowerPoint	Agent	SharePoint agent

The CW-401 training case

One chemical dosing skid links the day



One chemical dosing skid links the day

1 **Duty**

Deliver liquid consistently

2 **Concern**

Some later tests are low

3 **Evidence**

Test readings, failure review and weekly deviations

Start with an evidence packet

Copilot works better when the scope is small and explicit

1 Pump readings

12 time-ordered values

2 FMEA input

4 possible failure modes

3 Weekly data

6 improvement records

4 TR-401 extract

Fictional recording requirements

Copilot helps with the work around a decision

Use it to prepare evidence; keep approval with people

Copilot can

Summarise, calculate, format, compare and draft.

Engineer must

Confirm sources, assumptions, formulas, meaning and release.

A useful answer shows its limits

Three questions expose weak outputs

Source

Where did this claim come from?

Check

Can I reproduce the number?

Boundary

What does the evidence not prove?

Your two class agents

Ask them anything, all day, in your own words

IES Engineering Methods Tutor

Explains the three methods this course uses.

TRY ASKING

"Why is a high Detection score bad news?"

"What is the difference between percentage points and percent?"

"Why must control limits be frozen?"

Open from Copilot Chat → Agents,
or use the link in your lab folder.

IES AI Security Advisor

Answers safe and accountable AI use questions.

TRY ASKING

"Can I paste an NRIC into a prompt?"

"Who is accountable if the agent is wrong?"

"What happens to my prompt logs?"

Open from Copilot Chat → Agents,
or use the link in your lab folder.

Both agents are shared with the class. They teach and explain; they never approve an engineering decision.

Lab 1.1 · Onboarding and tool access check

Verify access once before the practical work

1 1

Sign in to Microsoft 365

2 2

Open Copilot Chat

3 3

Confirm Copilot in Word and Excel

4 4

Open Agents

5 5

Tell the trainer about any blocked surface

Lab 1.2 · AI versus Engineer challenge

Turn the same messy notes into a technical brief

Lab 1.2

Turn the same messy notes into a technical brief

ONE SMALL OUTPUT

1. **Input** — Later pump tests are low; strainer changed; verification also performed.
2. **Copilot team** — Create facts, assumptions and questions.
3. **Manual team** — Structure the same brief without Copilot.
4. **Compare** — Speed, completeness and unsupported claims.

Debrief · where was Copilot faster?

Productivity is useful only when the result survives review

Fast

Formatting, headings and possible questions

Needs review

Units, causation, requirements and sign-off

Module 1 recap

Choose the surface, provide evidence, check the draft

1 Choose

Match output to app

2 Ground

Use relevant sources

3 Verify

Open, calculate and challenge

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MODULE 2

Prompting for Engineering Documents

R.I.S.E. + E-MAP turn vague requests into reviewable work



Module 2 learning outcomes

By the end of this module, participants can:

1	**Explain**	Describe why prompt engineering improves productivity.
2	**Apply**	Build prompts with R.I.S.E. and E-MAP.
3	**Improve**	Add constraints, examples and output formats.
4	**Challenge**	Use multiple personas and a short debate to expose gaps.

Prompt engineering reduces avoidable rework

A clear request improves the first draft and the review

Faster

Fewer clarification cycles

Consistent

Repeatable structure

Traceable

Named sources and checks

Safer

Explicit boundaries and approval gates

Six principles for strong engineering prompts

Build enough structure for a useful and testable answer

1 Purpose

State the decision or output

2 Evidence

Name allowed sources

3 Context

Give system and units

4 Steps

Order the work

5 Constraints

Prevent harmful drift

6 Check

Define acceptance tests

R.I.S.E. · the practical prompt framework

Role · Input · Steps · Expectations

1 R · Role

Useful perspective

2 I · Input

Exact evidence and scope

3 S · Steps

Ordered work

4 E · Expectations

Format, limits and checks

R · Role sets perspective

A role guides attention; it does not grant authority

Useful

Act as an FMEA facilitator and preserve supplied ratings.

Unsafe

Act as the approving engineer and decide the system is safe.

I · Input names the evidence

Tell Copilot exactly what it may use

1 Source

Named table or document

2 Scope

Rows, date range and revision

3 Units

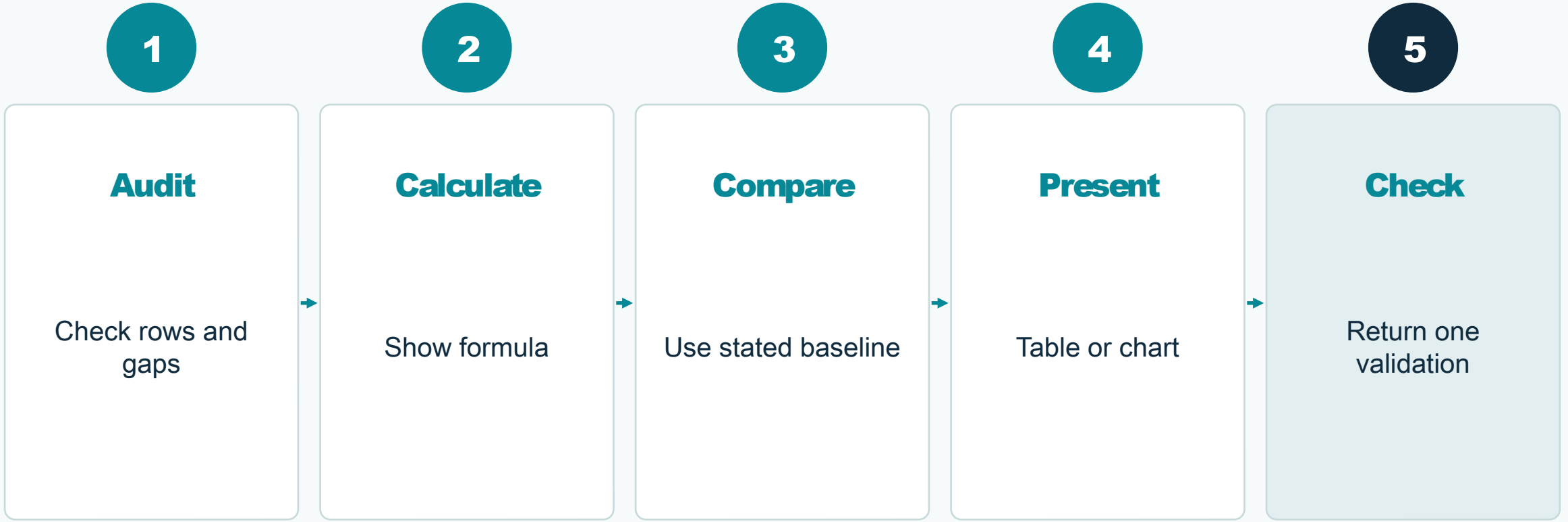
L/h, bar, minutes

4 Preserve

Do not alter source records

S · Steps make work traceable

Order calculations, comparisons and presentation



E · Expectations define “done”

Specify format, quality and decision boundary

1 Format

Five-row table; three-sentence summary

2 Evidence

Cite source row or clause

3 Limit

Do not infer root cause

4 Acceptance

Reproduce one number

Build one RISE prompt

Combine all four parts in a single request

PROMPT TO USE

Act as an engineering data assistant. Use only PumpReadings. Calculate the baseline mean and control limits, flag monitoring rows, create a chart, show formulas, and state what the chart cannot prove.

1. **Role**
   Engineering data assistant
2. **Input**
   Only PumpReadings
3. **Steps**
   Calculate, flag, chart
4. **Expected**
   Show formulas; no cause claim

Weak prompt versus R.I.S.E.

The same question, asked twice — and what changes in the answer

WEAK PROMPT

"Analyse this pump data and tell me if there is a problem."

What comes back

- No stated method, so you cannot repeat it
- Limits recalculated on the drifting data
- A confident cause, with no evidence
- No units, no formulas, nothing to check

R.I.S.E. PROMPT

"Act as an SPC analyst. Use only SPCData. Calculate limits from the 12 baseline subgroups, then freeze them. Show every formula."

What comes back

- A named method a reviewer can follow
- Limits from the baseline only, frozen
- A signal to investigate, not a cause
- Formulas and units you can check by hand

The difference is not politeness. Naming the evidence and the steps is what makes the answer reviewable — and what stops the model inventing the part you did not specify.

What E-MAP adds on top of R.I.S.E.

Engineering context, Metrics, Assumptions, Parameters

E	**Engineering context**	Without: "improve the dosing process"	With: "chemical dosing skid, weekly delivery test in L/h"
M	**Metrics**	Without: "show the improvement"	With: "deviation rate = deviations / opportunities, percent, 2 dp"
A	**Assumptions**	Without: assumptions stay hidden inside the answer	With: "state that the deviation definition was unchanged"
P	**Parameters**	Without: the model chooses its own scope	With: "six weeks only, three per phase, no other period"

E-MAP is what stops a generic answer. It pins the units, keeps the assumptions visible, and fences the data — so two engineers running the same prompt get the same result.

What better prompting actually buys you

Measured on the same task, by the same engineer

Rework

Fewer regeneration cycles because the format and scope were stated once.

Traceability

Every number carries a formula and a source, so review is fast.

Safety

The boundary is explicit, so the model does not approve or diagnose.

Repeatability

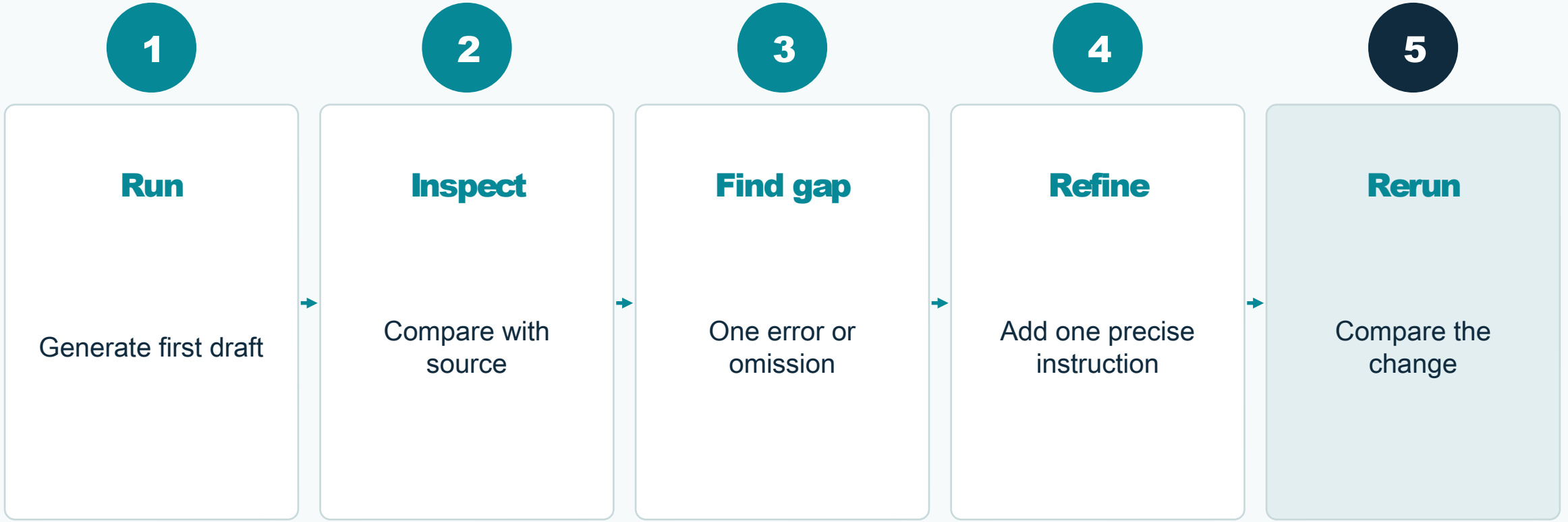
A colleague can rerun the prompt and get a comparable answer.

The one habit that matters most

Name the evidence the model may use, and forbid everything else. Almost every invented failure mode, phantom clause and confident wrong cause in this course traces back to a prompt that did not say which data was in scope.

Iterative prompting is a review loop

Check one gap, refine one instruction, rerun



E-MAP adds engineering context

Engineering · Metrics · Assumptions · Parameters

1 E · Engineering

System, function, lifecycle

2 M · Metrics

Units, formula, target

3 A · Assumptions

Known, unknown, provisional

4 P · Parameters

Limits, conditions, revision

Engineering context prevents generic answers

Describe the system and the decision



1 **System**

CW-401 chemical dosing skid

2 **Function**

Deliver liquid at a controlled rate

3 **Lifecycle**

Standardised delivery test

4 **Decision**

Investigate unusual change

Metrics make comparison explicit

Name numerator, denominator, units and time basis

Flow rate

Volume × 3600 ÷ duration

Deviation rate

Deviations ÷ opportunities

RPN

Severity × Occurrence × Detection

Assumptions must remain visible

Separate what is known from what is provisional

Known

T10 = 59.4 L/h

Assumption

Test conditions were comparable

Unknown

Which intervention changed performance

Parameters define the operating boundary

Give limits, revisions and conditions

1 Revision

TR-401 revision 1

2 Reference

Eight Baseline readings

3 Specification

58.9–65.1 L/h fictional range

4 Rule

Do not change source rows

Combine RISE and E-MAP

RISE structures the request; E-MAP supplies technical detail

1

RISE

Who, evidence, work, output

2

E-MAP

System, measures, assumptions, limits

3

Result

A draft that can be checked

Constraints make the answer reviewable

State what must be preserved and what must not be claimed

Data

Keep units, order and missing values

Formula

Show rows and parameters

Evidence

Use named sources only

Decision

Do not invent cause or approval

Weak and strong constraints

Precise boundaries are easier to test

Weak

Be accurate and careful.

Strong

Use only FMEAInput; preserve four rows and ratings; show RPN formula; flag Severity ≥ 9; do not approve risk.

Request a fixed output schema

A small structure makes review faster

1 Observation

What changed?

2 Evidence

Which value supports it?

3 Limitation

What remains unknown?

4 Next check

What evidence is needed?

Ask Copilot to expose its evidence

Traceability begins inside the prompt

1 Source

Name sheet, table or clause

2 Formula

Show calculation

3 Assumption

Label provisional statements

4 Citation

Provide a link or passage

5 Unknown

Say when evidence is absent

Multi-persona review catches different gaps

Use viewpoints that have distinct responsibilities

Reliability

Failure mechanisms and maintainability

Operations

Conditions, sequence and practicality

Safety

Hazards and safeguards

Quality

Evidence, criteria and traceability

Multi-persona prompt example

Ask each role to challenge the same draft

PROMPT TO USE

Review the draft from three perspectives: reliability, operations and safety. Each persona must identify one unsupported claim, one missing piece of evidence and one practical check. Preserve disagreements.

1. **Reliability**
   What failure mechanism is unsupported?
2. **Operations**
   Which condition may have changed?
3. **Safety**
   What must be checked before action?

Let the personas debate before synthesis

Disagreement can reveal assumptions hidden by consensus

Claim	Reliability	Operations	Safety	Synthesis
“The strainer replacement solved the problem.”	Was blockage confirmed?	Two interventions happened together.	What must be checked before changing controls?	Performance improved; cause remains unisolated.

Debate prompt example

Require argument, rebuttal and a bounded conclusion

PROMPT TO USE

Run a two-round review. Reliability, Operations and Safety each state one claim with evidence. In round two, each challenges one other claim. Finish with agreed facts, unresolved disagreements and the smallest test that could resolve them. Do not force consensus.

1 **Round 1**
Each persona gives evidence

2 **Round 2**
Challenge another claim

3 **Final**
List agreement, disagreement and next test

The scenario behind Labs 2.1 to 2.4

One roadside drainage inspection, carried across four Copilot apps

1

The job

A worker inspects roadside drains on the public road network

2

The rule

Silt depth must stay at or under 50 mm

3

One passes

WO-021 at DR-17 reads 32 mm and goes to approval

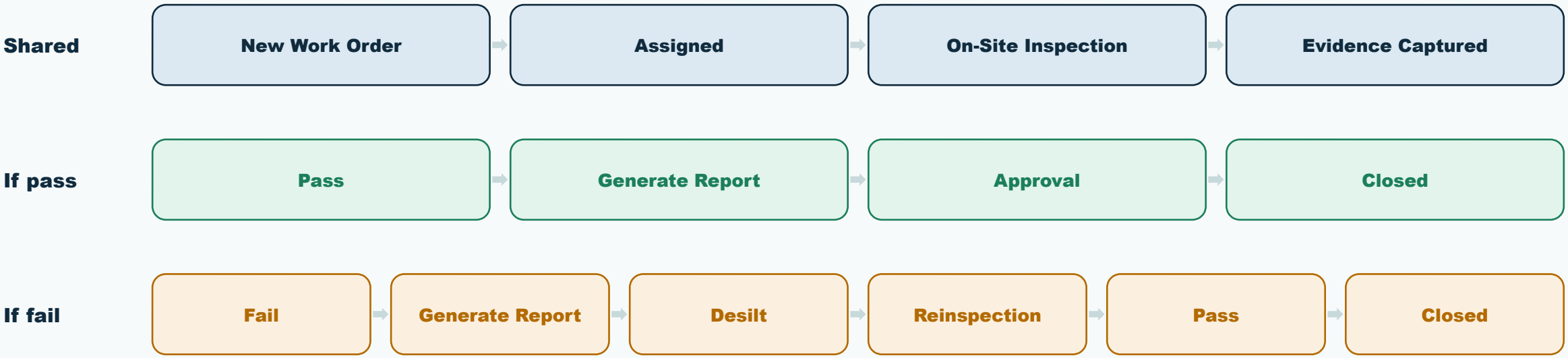
4

One fails

WO-022 at DR-23 reads 74 mm, is desilted, then passes

The work order workflow

Every lab in Module 2 sits somewhere on this path



WO-021 takes the green path and WO-022 the amber one. Labs 2.1 and 2.2 build the path; Lab 2.3 requests the approval; Lab 2.4 reports the week.

Lab 2.1 · Work order intake and assignment

Excel · two drainage work orders, from intake to the Pass/Fail decision

1

Excel

The register, already a named table

2

Status history

Six ordered states, one row per change

3

Decision

32 mm passes, 74 mm fails against 50 mm

4

Output

Work-Order-Register.xlsx, carried forward

Lab 2.2 · Inspection reports and rectification

Word · one order passes, one fails, is desilted and reinspected

1

Word

Two reports drafted from the register

2

Margin

+18, -24 and +22 mm, checked by hand

3

Fail branch

74 mm fails, desilted, 28 mm passes

4

Versioning

RPT-022-v2 keeps the first failure visible

Lab 2.3 · Email the report to the manager

Outlook · request approval to close, without claiming you have it

1

Outlook

Two emails drafted with Copilot

2

The gate

Approval is requested, never assumed

3

The risk

A summary that hides the initial failure

4

Output

Approval_Log records what was sent, when

5

Review

Every figure traced back to the source

Lab 2.4 · Weekly update to the supervisor

PowerPoint · six inspections, one week, five slides

1	2	3	4	5
PowerPoint	**The week**	**Pass rate**	**Still open**	**Review**
The deck built from your own register	6 inspections, 3 failed first visit	50.0%, worked out by hand first	WO-024 at DR-38, 88 mm, not rectified	Every figure traced back to the source

PowerPoint · create a management summary

Communicate the checked decision, evidence and next action

1

Headline

Observed change

2

Evidence

One chart and two values

3

Boundary

Cause not yet proven

4

Decision

Controlled next test

Module 2 recap

Prompt quality is visible in the result

1 Structure

RISE

2 Technical context

E-MAP

3 Improve

Constraints + personas + iteration

4 Accept

Evidence + independent check

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MODULE 3

Engineering Methodology Primer & Document Automation

Understand FMEA → DMAIC → SPC before asking Copilot to build the outputs



Module 3 learning outcomes

By the end of this module, participants can:

1	**FMEA**	Explain failure mode, effect, cause, control and RPN.
2	**DMAIC**	Explain five phases and three everyday analysis tools.
3	**SPC**	Explain control charts, variation and Cp/Cpk.
4	**Apply**	Use Copilot in Excel and verify one result per method.

Use the methods in this sequence

Risk → improvement → monitoring, and that is the order of your three labs

1

1 · FMEA

What could fail? · Lab 3.1

2

2 · DMAIC

How will we improve? · Lab 3.2

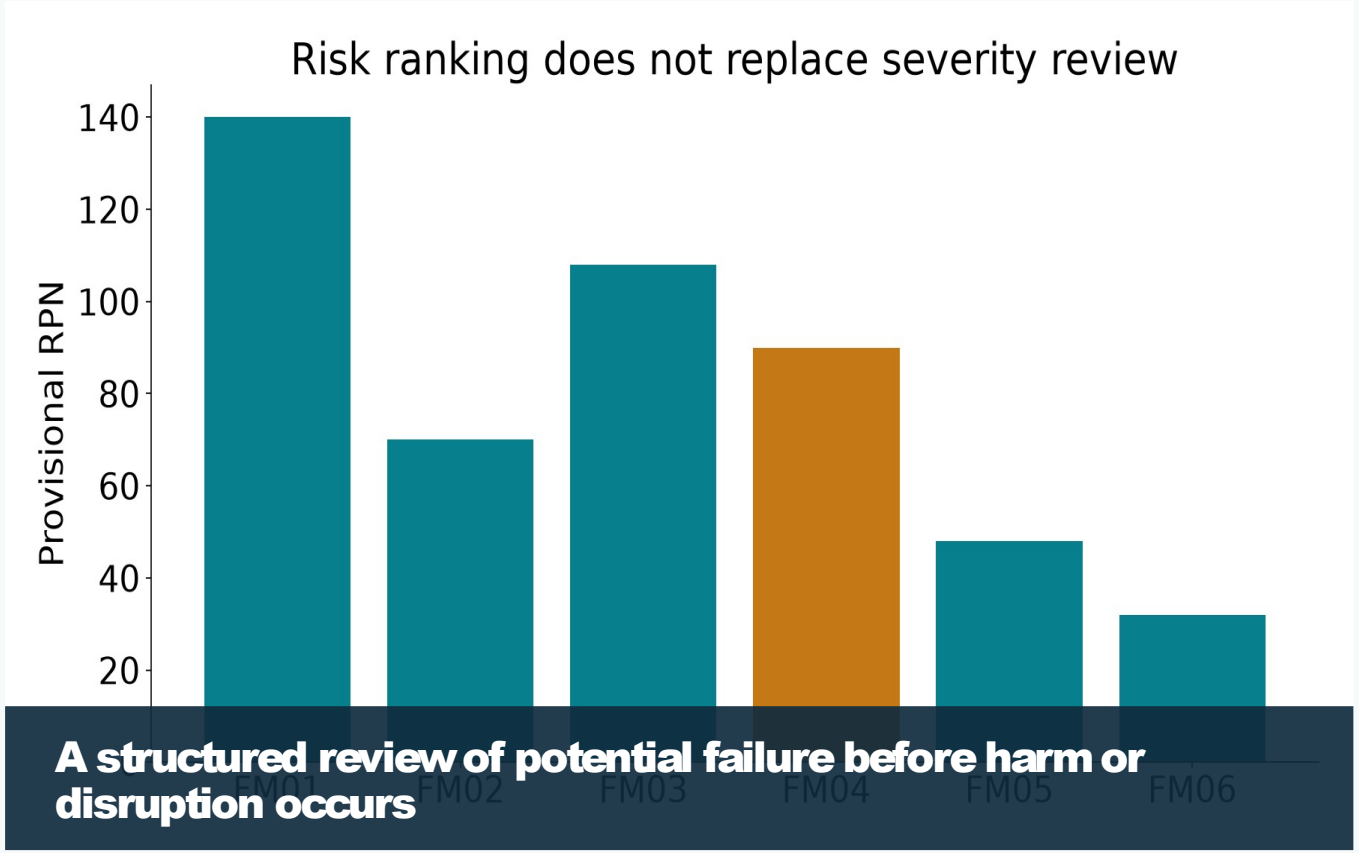
3

3 · SPC

Is the process stable? · Lab 3.3

FMEA · what it is

A structured review of potential failure before harm or disruption occurs



1 Full name

Failure Mode and Effects Analysis

2 Question

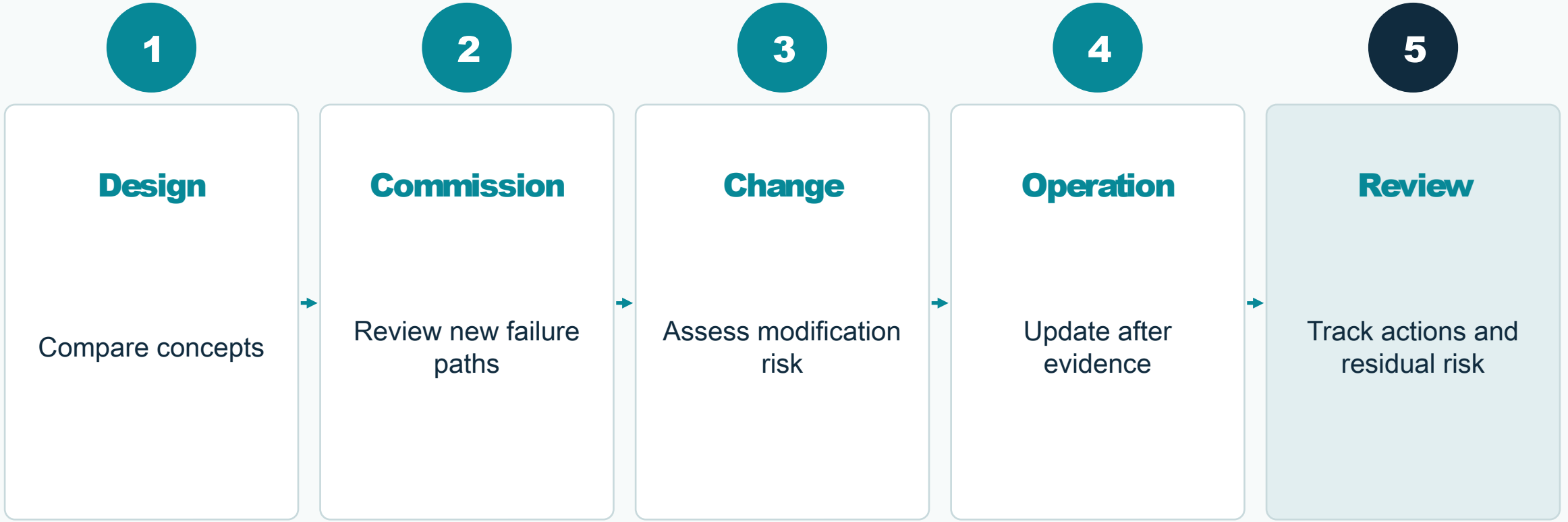
What could fail, with what effect, and how is it controlled?

3 Output

Prioritised action list with evidence gaps

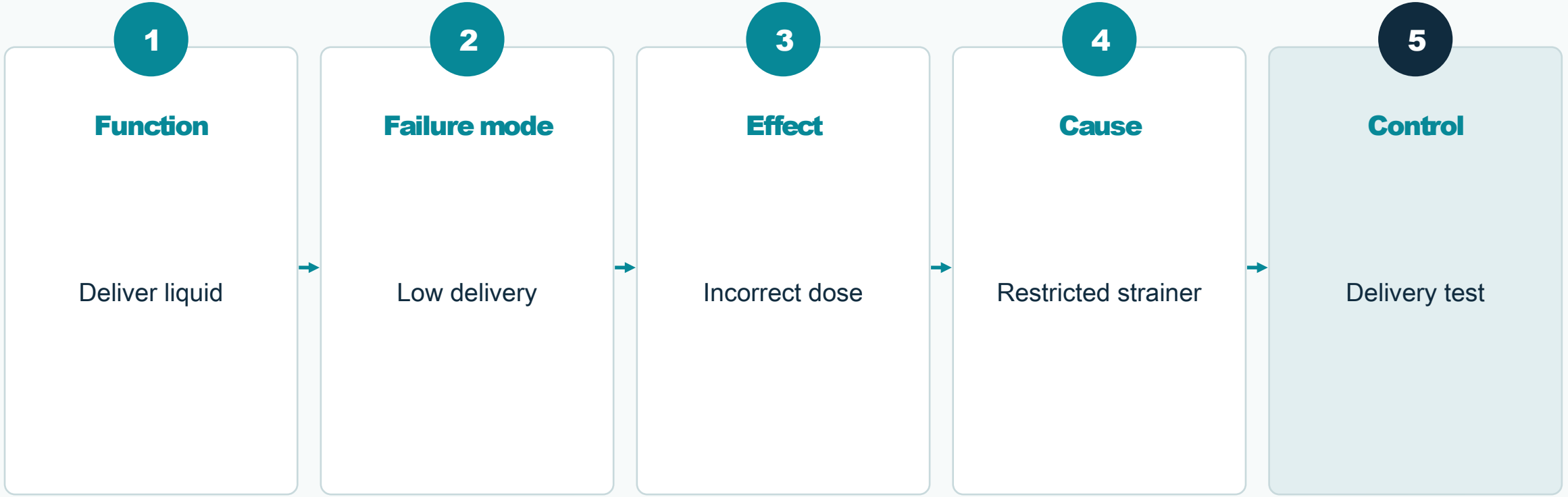
When engineers use FMEA

Use it before release and when systems or processes change



An FMEA row tells one risk story

Read the columns from left to right



Failure mode, effect, cause and control

Four terms answer four different questions

Failure mode	Effect	Cause	Control
How could the function fail?	What happens next?	Why might it happen?	How is it prevented or detected?

Severity, Occurrence and Detection

Three ratings support prioritisation

1 Severity

How serious is the effect?

2 Occurrence

How likely is the cause?

3 Detection

How unlikely are current controls to detect it? Higher D = poorer detection.

RPN makes the calculation visible

Risk Priority Number = Severity × Occurrence × Detection

FM01	RPN	Meaning
7 × 5 × 4	140	Priority for review, not probability

A lower RPN can still need urgent action

Use severity gates and engineering judgement

FM01

RPN 140 · Severity 7

FM04

RPN 90 · Severity 9

Decision

Review FM04 despite lower RPN

Worked FMEA · four supplied rows

Copilot calculates; the team owns ratings and actions

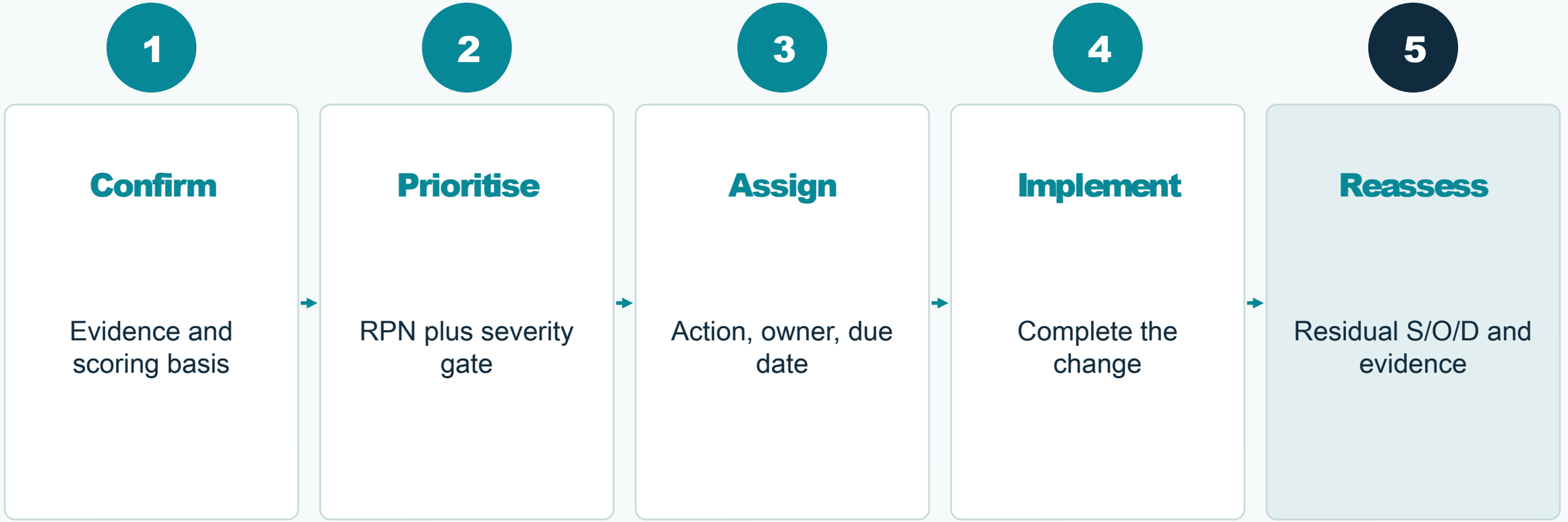
ID	Failure mode	S	O	D	RPN
FM01	Low delivery	7	5	4	**140**
FM02	No start	8	3	3	**72**
FM03	Wrong reading	6	4	5	**120**
FM04	Leak	9	2	5	**90**

How to act

1. Check the rating basis
2. Rank RPNs
3. Apply severity gate
4. Assign action and owner
5. Reassess residual risk

Act on an FMEA result

Prioritisation should lead to owned, verified work



How Copilot improves FMEA productivity

Reduce table setup and drafting time

Copilot helps

Draft rows, calculate RPN, sort, flag and format.

Engineer verifies

System scope, failure logic, ratings, actions and residual risk.

FMEA prompt · preserve the engineering record

Name the table, formula, gate and decision boundary

PROMPT TO USE

Act as an FMEA facilitator. Use only FMEAInput. Preserve all rows and supplied ratings. Calculate RPN = Severity × Occurrence × Detection, rank the rows, flag Severity 9 or higher, and add one evidence request and action per row. Show formulas. Do not invent incidents or approve the design.

1 **Input**
FMEAInput · 4 rows

2 **Output**
RPN, rank and action

3 **Check**
FM01 = 140; FM04 S = 9

Lab 3.1 · FMEA document and RPN table

20 minutes · one small table and two independent checks

Lab 3.1

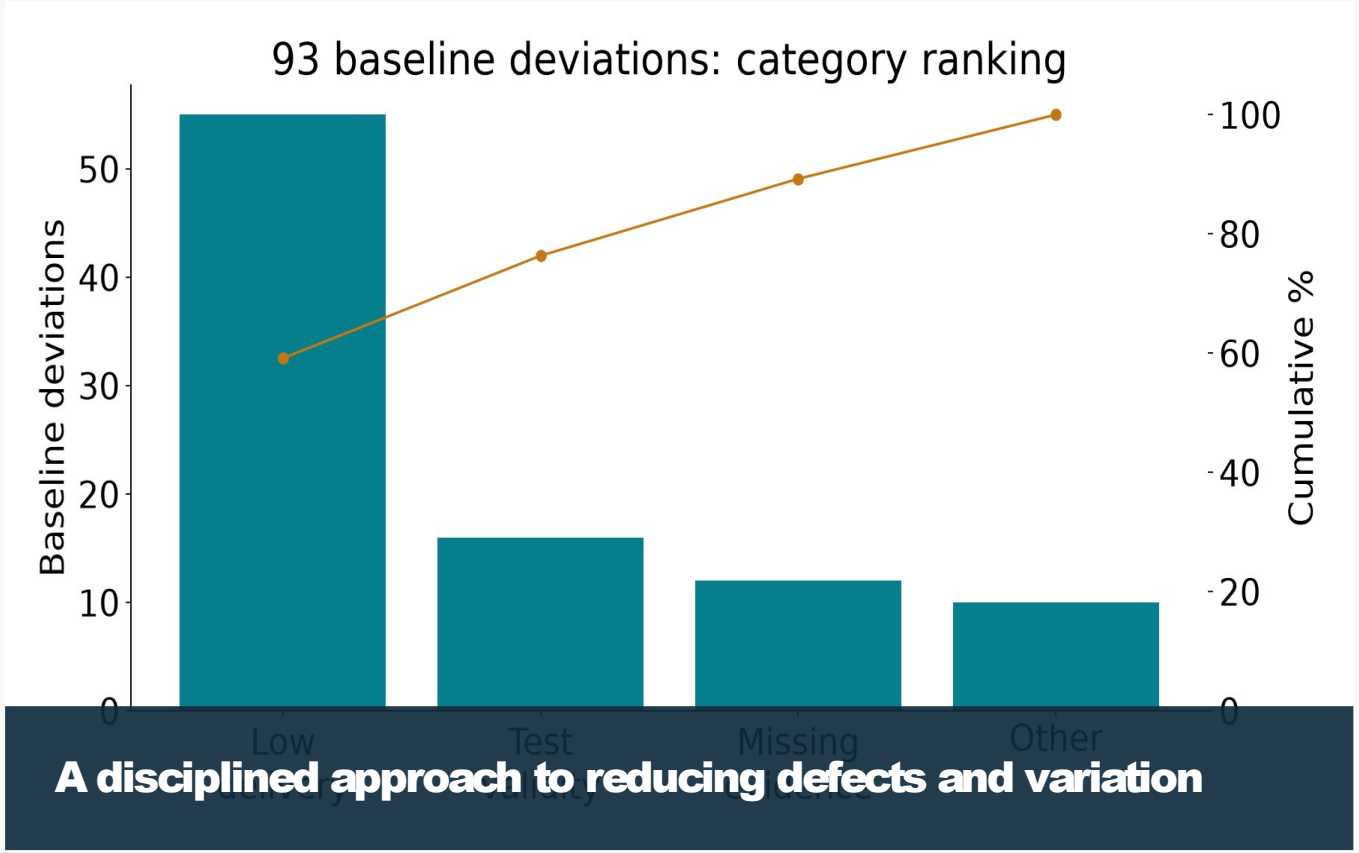
20 minutes · one small table and two independent checks

ONE SMALL OUTPUT

1	Open	Labs/Lab-3.1-FMEA-Document/Starter.xlsx
2	Select	FMEAInput
3	Ask	Paste Copilot_Prompt
4	Verify	FM01 = 140; FM04 Severity = 9
5	Save	Ranked table + evidence request

Six Sigma · what it is

A disciplined approach to reducing defects and variation



1. **Focus**

   Customer-critical performance

2. **Evidence**

   Measured process behaviour

3. **Method**

   DMAIC for improving an existing process

DMAIC is the improvement roadmap

Define · Measure · Analyse · Improve · Control



Define · make the problem measurable

State what, where, when and impact

1 Problem

Delivery-test deviations occur

2 Scope

CW-401 standardised tests

3 Metric

Deviations ÷ opportunities

4 Boundary

No cause claim yet

Measure · establish a trustworthy baseline

Confirm definitions before calculating

1 Data definition

What counts as a deviation?

2 Denominator

How many opportunities?

3 Quality

Missing, duplicate or invalid rows?

4 Baseline

Which period represents current performance?

Analyse · separate pattern from cause

Use tools to generate and test explanations

1

Pareto

Which categories dominate?

2

Fishbone

What possible causes exist?

3

5-Whys

What causal chain should be tested?

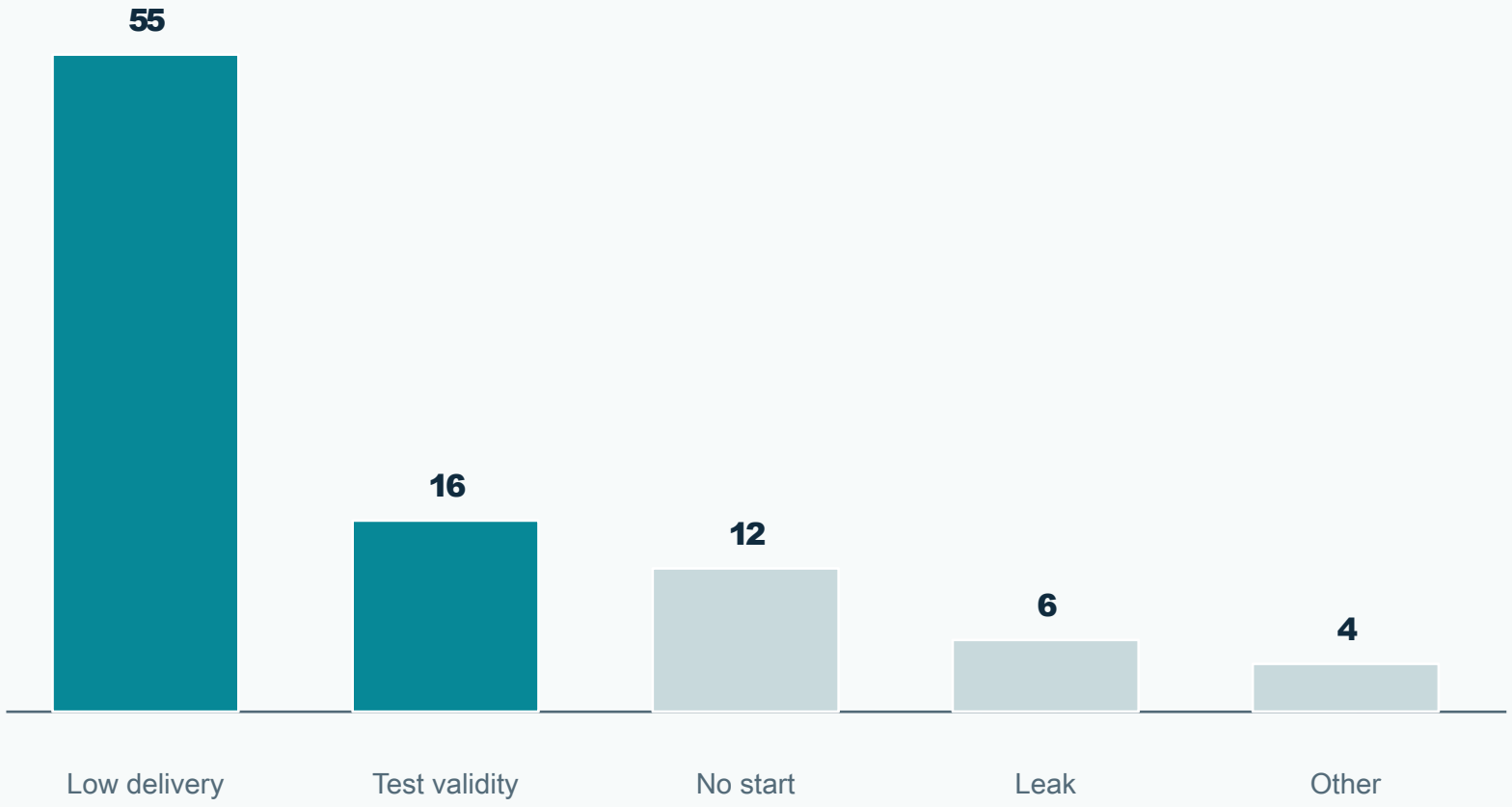
4

Evidence

Which records or tests can confirm it?

Pareto · focus on the largest categories

Bars descend; the cumulative line shows contribution



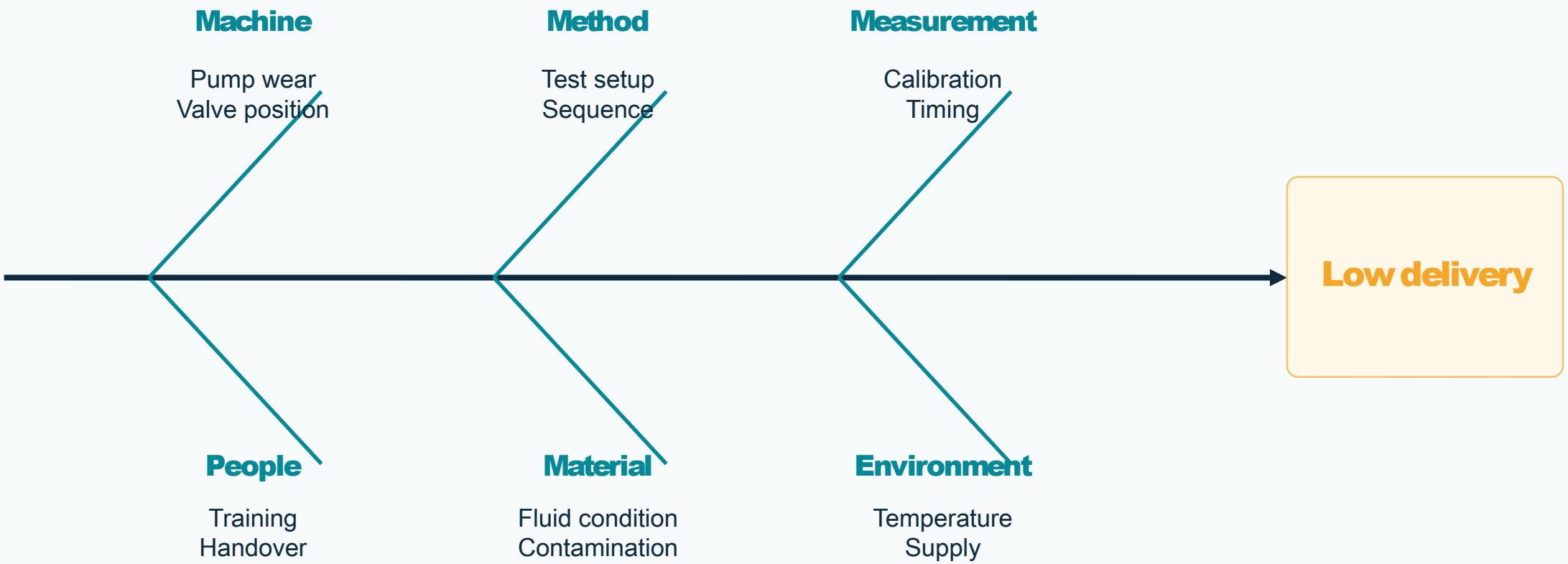
Read it correctly

Largest bars show where recorded problems concentrate.

They do not prove root cause.

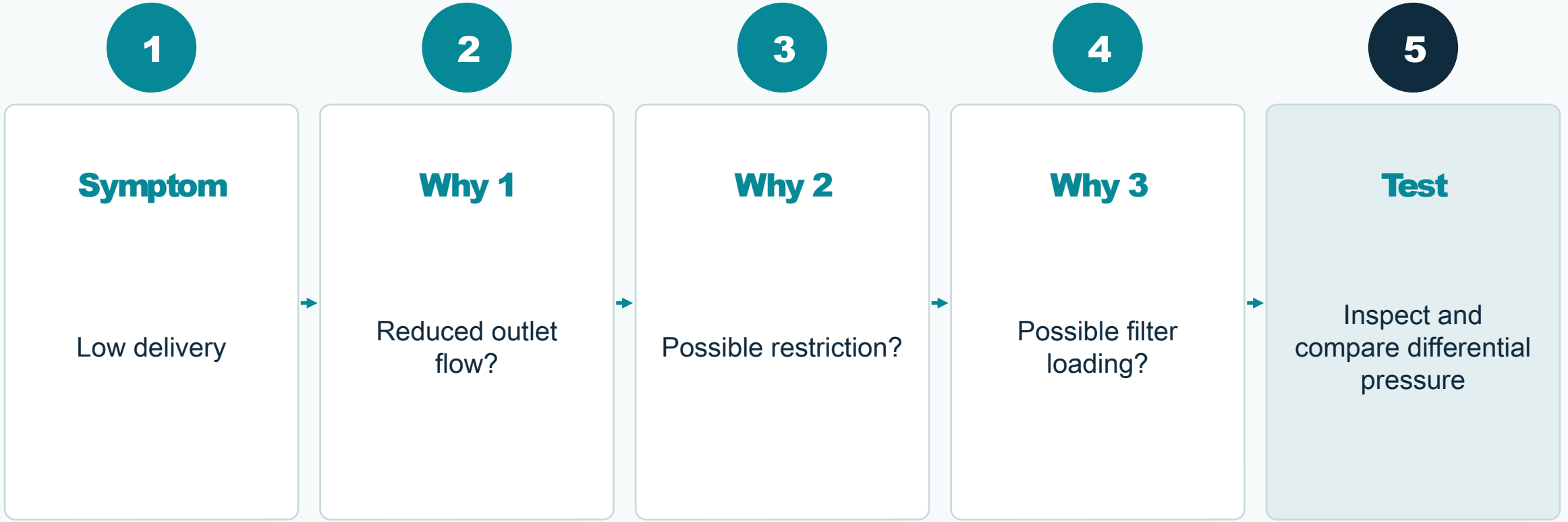
Fishbone · organise possible causes

Explore broadly before choosing what to test



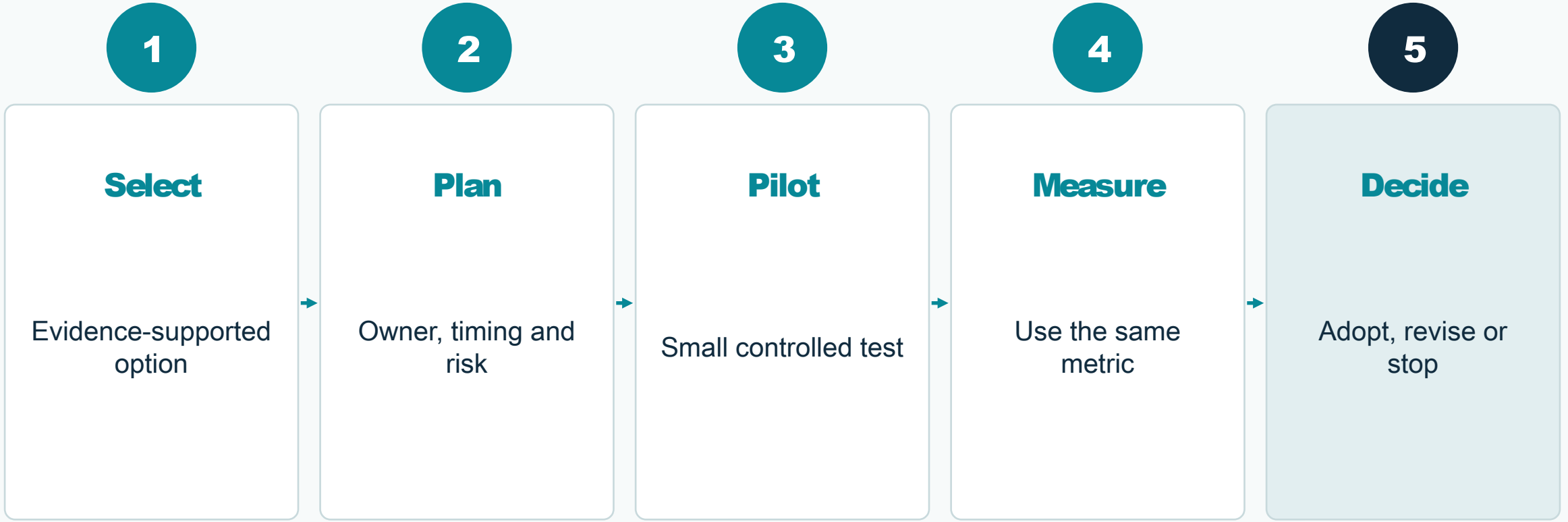
5-Whys · build a testable causal chain

Stop when evidence is missing



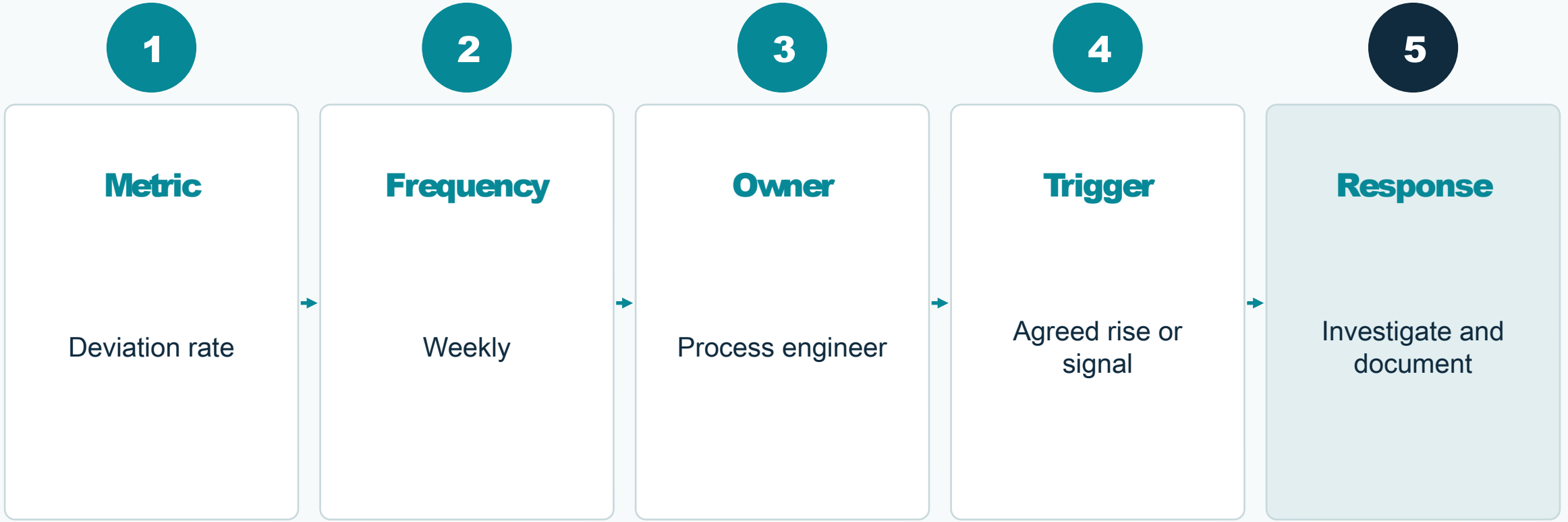
Improve · test a controlled change

Select an action that addresses supported causes



Control · sustain the gain

Turn the result into a monitoring and response plan



Worked DMAIC comparison

Rates make unequal periods comparable

Baseline	Pilot	Change	Boundary
36 ÷ 400 = 9.0%	7 ÷ 200 = 3.5%	−5.5 percentage points	Short pilot does not prove cause

How Copilot improves DMAIC productivity

Draft phase summaries while preserving evidence gaps

Copilot helps

Profile data, calculate rates, suggest charts and draft five phases.

Engineer verifies

Metric definition, cause evidence, test design and control ownership.

DMAIC prompt · facts before hypotheses

Request five phases and an explicit limitation

PROMPT TO USE

Act as a Six Sigma project assistant. Use only WeeklyData. Calculate Baseline and Pilot deviation rates. Create a five-row Define, Measure, Analyse, Improve, Control summary. Separate facts from hypotheses, show formulas, and state that the short pilot does not prove cause.

1 **Input**
WeeklyData · 6 rows

2 **Output**
Two rates + five-row DMAIC

3 **Check**
9.0% and 3.5%

Lab 3.2 · DMAIC improvement report

30 minutes · phase rates, Pareto evidence and a five-phase summary

Lab 3.2

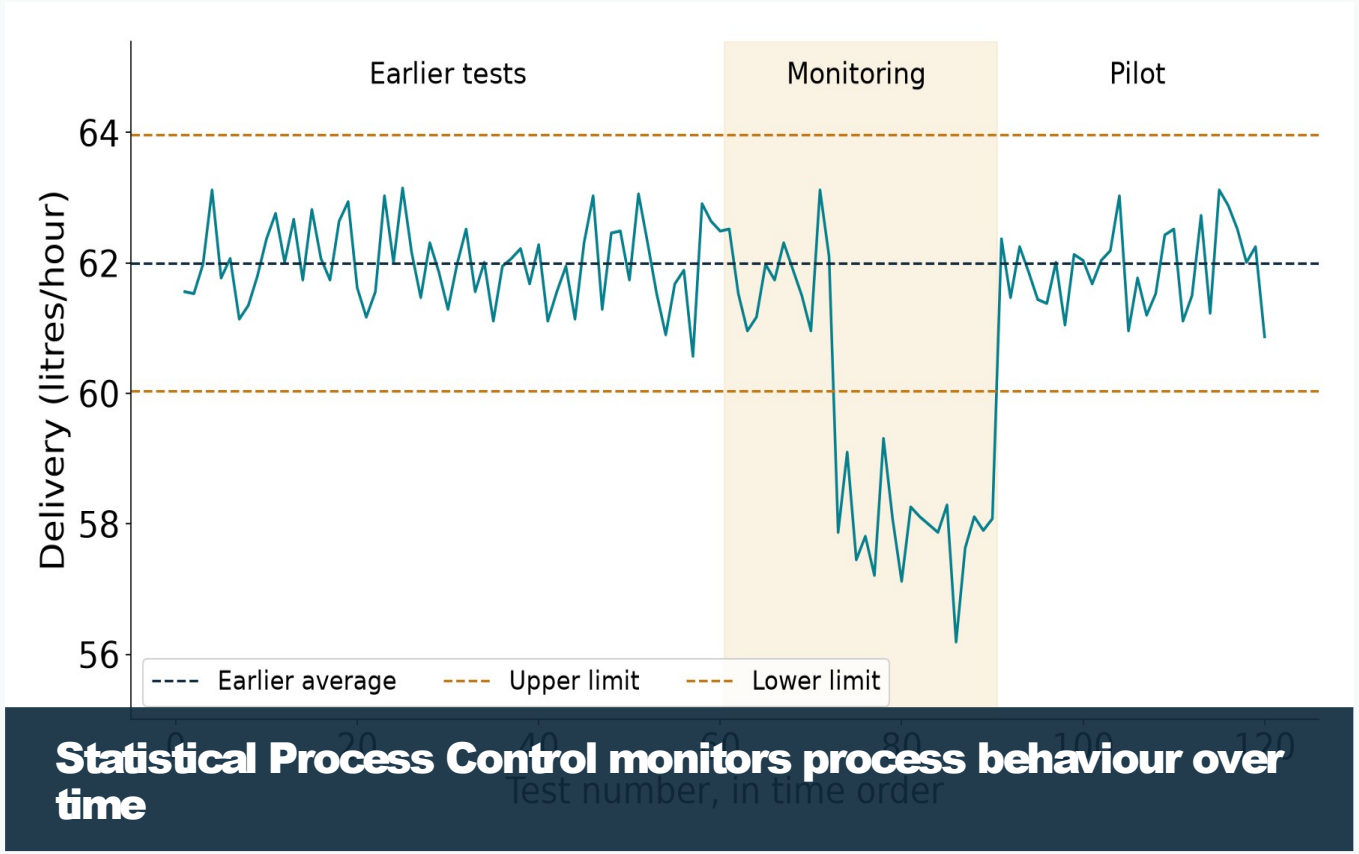
30 minutes · phase rates, Pareto evidence and a five-phase summary

ONE SMALL OUTPUT

1	**Open**	Lab-3.2-DMAIC-Report/Starter.xlsx
2	**Select**	WeeklyData
3	**Ask**	Paste the RISE + E-MAP prompt
4	**Verify**	36/400 = 9.00%; 7/200 = 3.50%
5	**Save**	Five phases + Pareto + next evidence check

SPC - what it is

Statistical Process Control monitors process behaviour over time



Statistical Process Control monitors process behaviour over time

1 **Use when**

Repeated measurements exist

2 **Question**

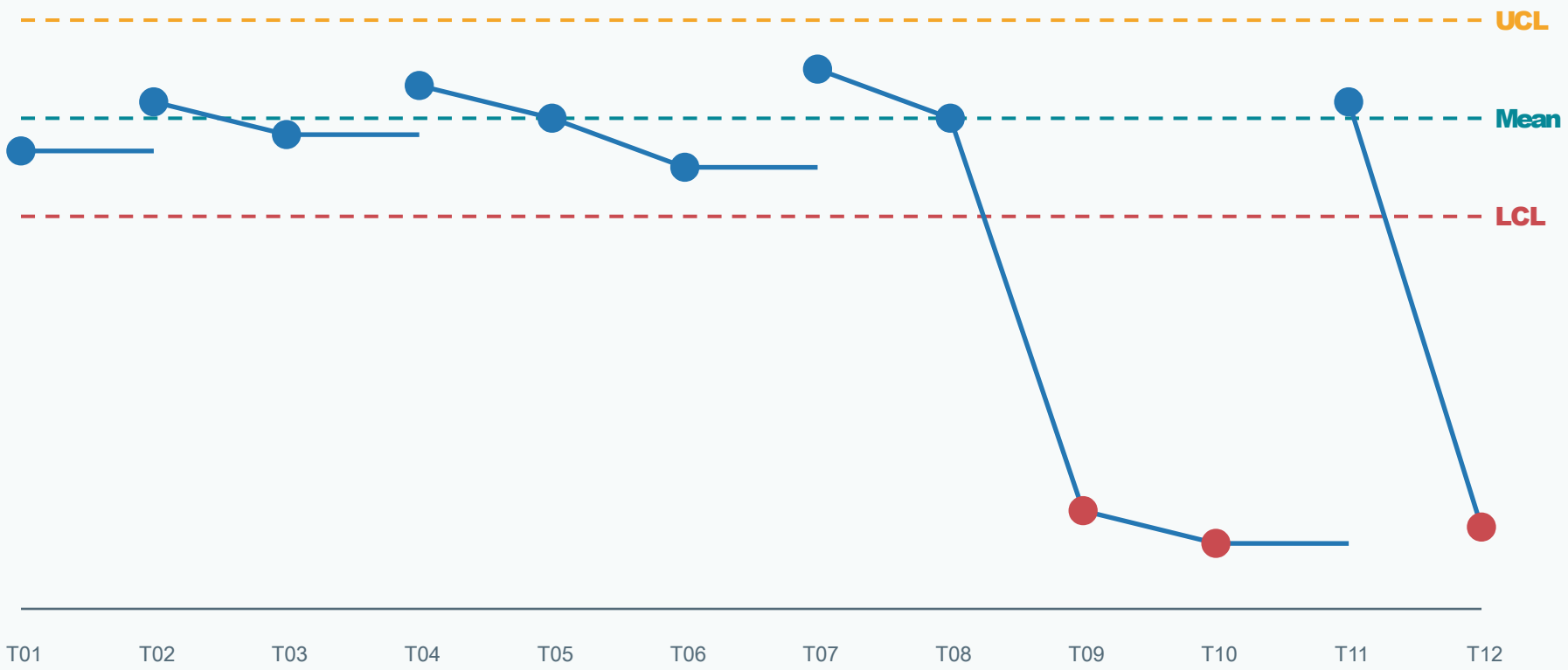
Stable pattern or unusual signal?

3 **Output**

Control chart and investigation trigger

A control chart has four essential parts

Time order is what makes the pattern meaningful



Read left → right

Baseline
T01–T08

Monitoring
T09–T12

Signals
T09, T10, T12

Common cause and special cause

Separate routine variation from a signal worth investigating

Common cause

Small fluctuations produced by the current process system.

Special cause

An unusual point or pattern indicating something changed.

Read signals with agreed rules

One point beyond a limit is the simplest rule

1 Point beyond UCL/LCL

Immediate investigation signal

2 Long run on one side

Possible shift

3 Sustained trend

Possible drift

4 Repeating pattern

Possible cyclic influence

Control limits are not specification limits

Behaviour and requirement answer different questions

Control limits

Calculated from process data; show expected behaviour.

Specification limits

Set by design, customer or regulation; show acceptable result.

Example

59.5 L/h meets 58.9–65.1 spec but is below the 61.4 LCL.

Cp and Cpk in plain terms

Capability compares stable process spread with specifications

Cp

Could the spread fit if centred?

Cpk

Does it fit where the mean actually sits?

> 1

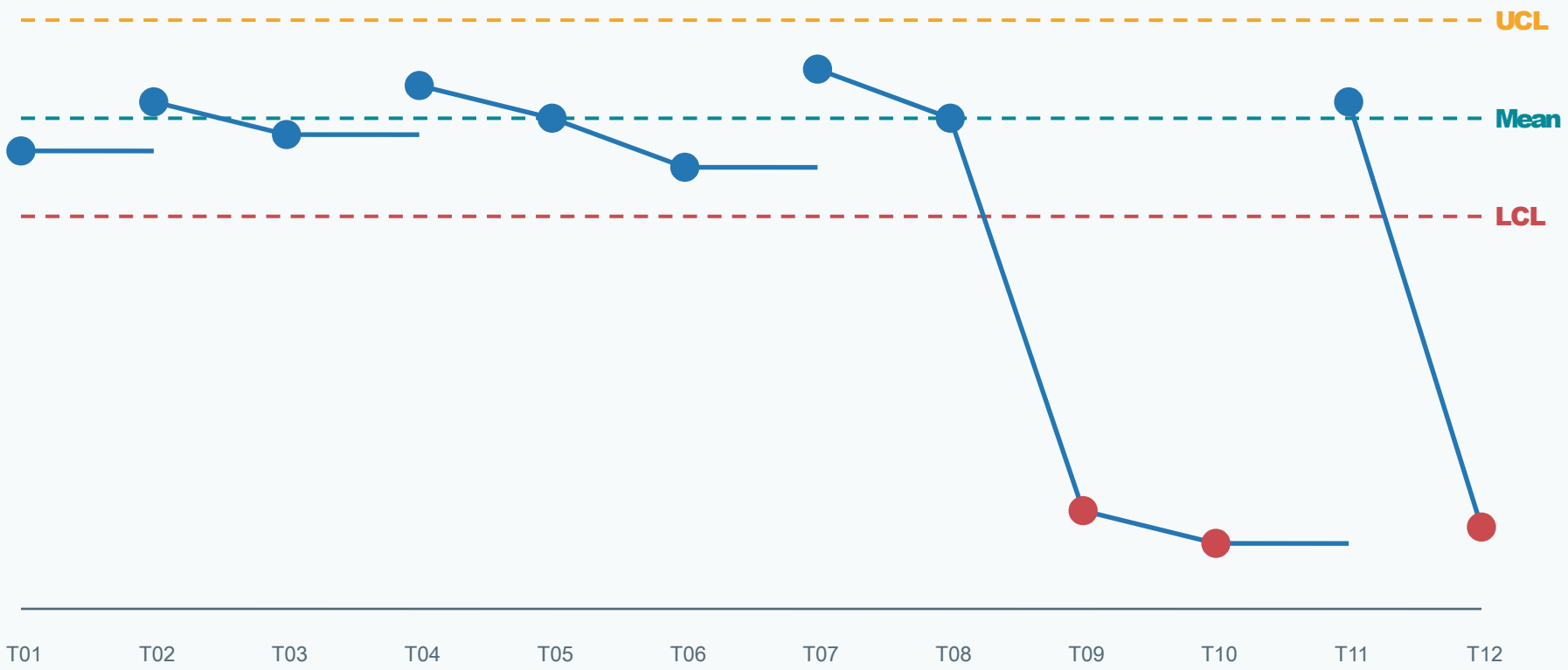
Estimated spread fits inside limits

Check first

Stability, data quality and distribution

Worked SPC interpretation

Three low monitoring points are signals to investigate



Read left → right

Baseline
T01–T08

Monitoring
T09–T12

Signals
T09, T10, T12

How Copilot improves SPC productivity

Speed up formulas, flags and chart preparation

Copilot helps

Calculate limits, create helper columns, flag rows and draft interpretation.

Engineer verifies

Baseline choice, formulas, signals, specification and process context.

SPC prompt · make the baseline explicit

State the calculation, chart and evidence boundary

PROMPT TO USE

Act as an engineering data assistant. Use only PumpReadings. From the eight Baseline rows, calculate the mean, sample standard deviation, UCL and LCL. Flag T09–T12, create a line chart with fixed limits, show formulas, and give a three-sentence interpretation. Do not claim a root cause or safety conclusion.

1 **Input**
PumpReadings · 12 rows

2 **Output**
Limits, flags and chart

3 **Check**
T10 against LCL

Lab 3.3 · SPC control chart and capability

30 minutes · Xbar-R chart from a frozen baseline, then Cp and Cpk

Lab 3.3

30 minutes · Xbar-R chart from a frozen baseline, then Cp and Cpk

ONE SMALL OUTPUT

1	**Open**	Lab-3.3-SPC-Control-Chart/Starter.xlsx
2	**Select**	SPCData · n = 4
3	**Ask**	Limits from subgroups 1-12 only, then freeze
4	**Verify**	Subgroup 13 Xbar = 61.275 < LCL 61.547
5	**Read**	Cp >> Cpk means off centre, not incapable

IES

MODULE 4

Copilot Agents & Microsoft 365 Reasoning Agents

Automate a controlled multi-step workflow



Module 4 learning outcomes

By the end of this module, participants can:

1	**Explain**	Recognise when a reusable agent is justified.
2	**Build**	Configure instructions and knowledge for an FMEA Assistant.
3	**Test**	Run complete and missing-data cases.
4	**Apply**	Use Researcher and Analyst with independent checks.

An agent packages a repeatable job

Instructions + knowledge + tools + conversation

1

Instructions

Role, steps and limits

2

Knowledge

Templates and references

3

Actions

Work the agent can perform

4

Tests

Expected behaviours and failures

Build an agent when the task repeats

Use a simple decision test

Repeatable?

Same inputs and output structure

Bounded?

Clear evidence and limits

Checkable?

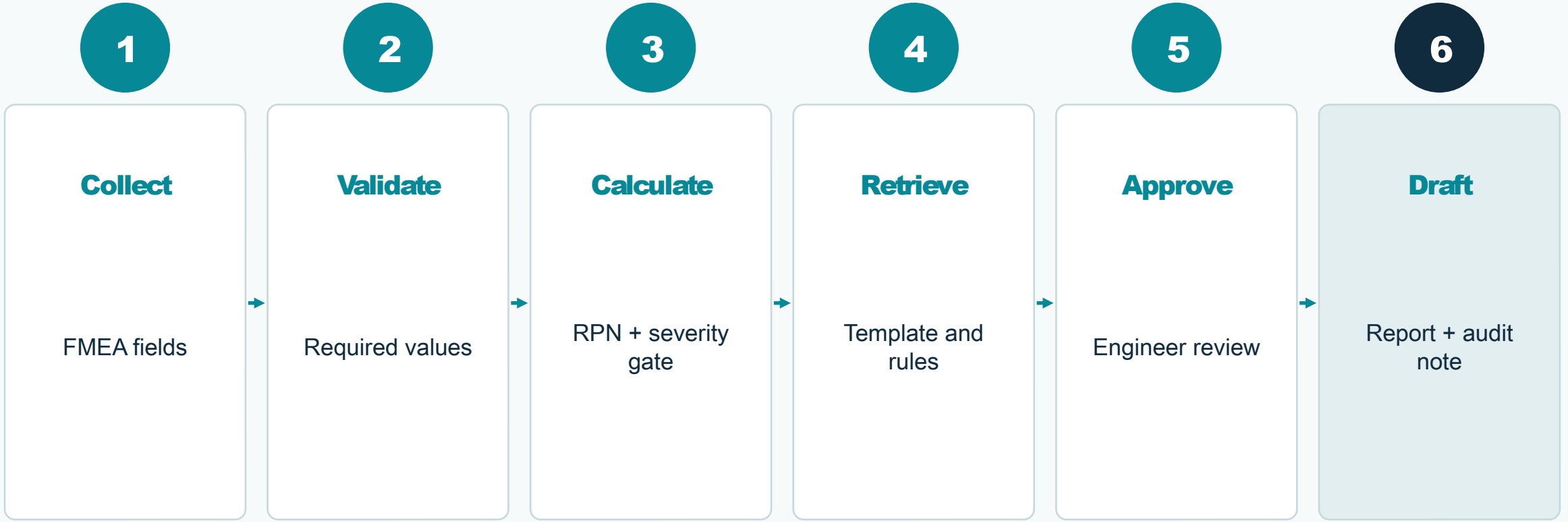
Acceptance tests exist

Governed?

Audience and permissions known

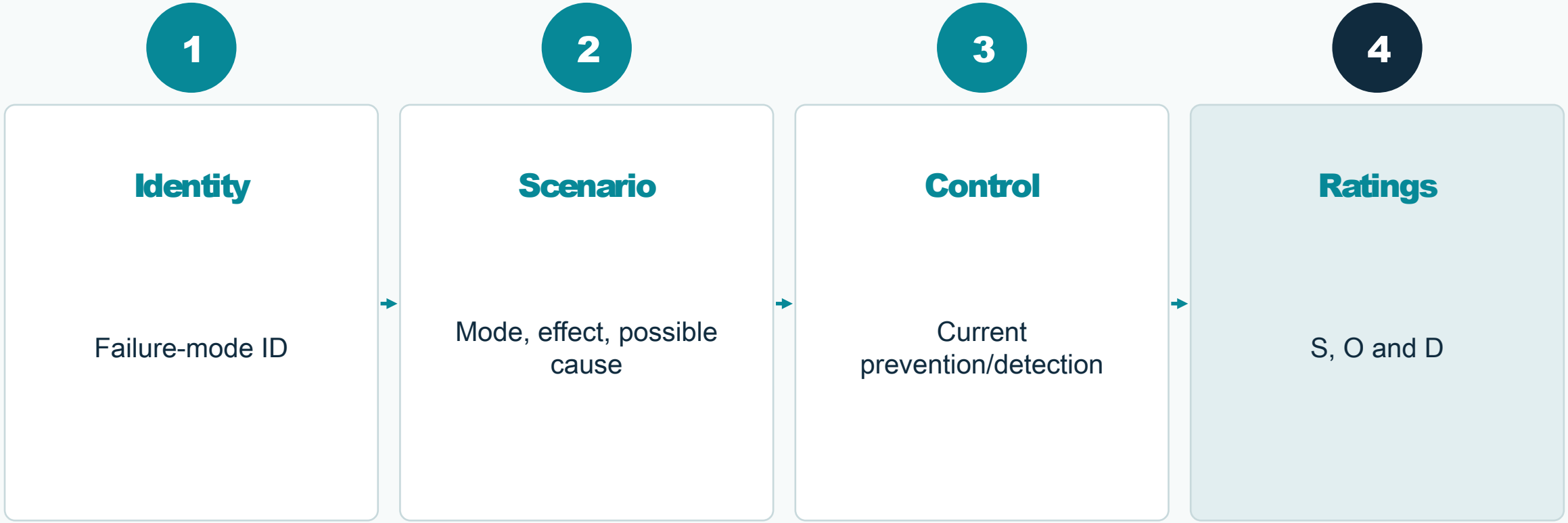
Agent workflow · FMEA to report

Six controlled steps with a human gate



Step 1 · collect only required fields

A small input contract exposes missing information



Step 2 · validate before calculating

The agent should stop when critical data is missing

All S/O/D present

Continue to calculation

Detection missing

Ask for value and scoring basis

Source unclear

Request revision or record

Conflicting values

Show conflict; do not choose silently

Step 3 · calculate visibly

Show source values and arithmetic

Inputs

S = 7 · O = 5 · D = 4

Formula

7 × 5 × 4

Result

RPN = 140

Step 4 · retrieve the right template

Ground the output format on a controlled source

1

Knowledge

FMEA report template

2

Version

Confirm revision

3

Mandatory

Evidence gap, action, owner, status

4

Prohibited

No invented approval

Step 5 · place a human approval gate

No report is released before engineering review

Agent output	Engineer review	Approver	Decision
Draft only	Logic, ratings and evidence	Named competent role	Approve, revise or reject

Step 6 · create a traceable report

Return the work needed for review and audit

1 Input record

Source fields

2 Calculation

RPN and gate

3 Draft action

Owner and due date

4 Limitations

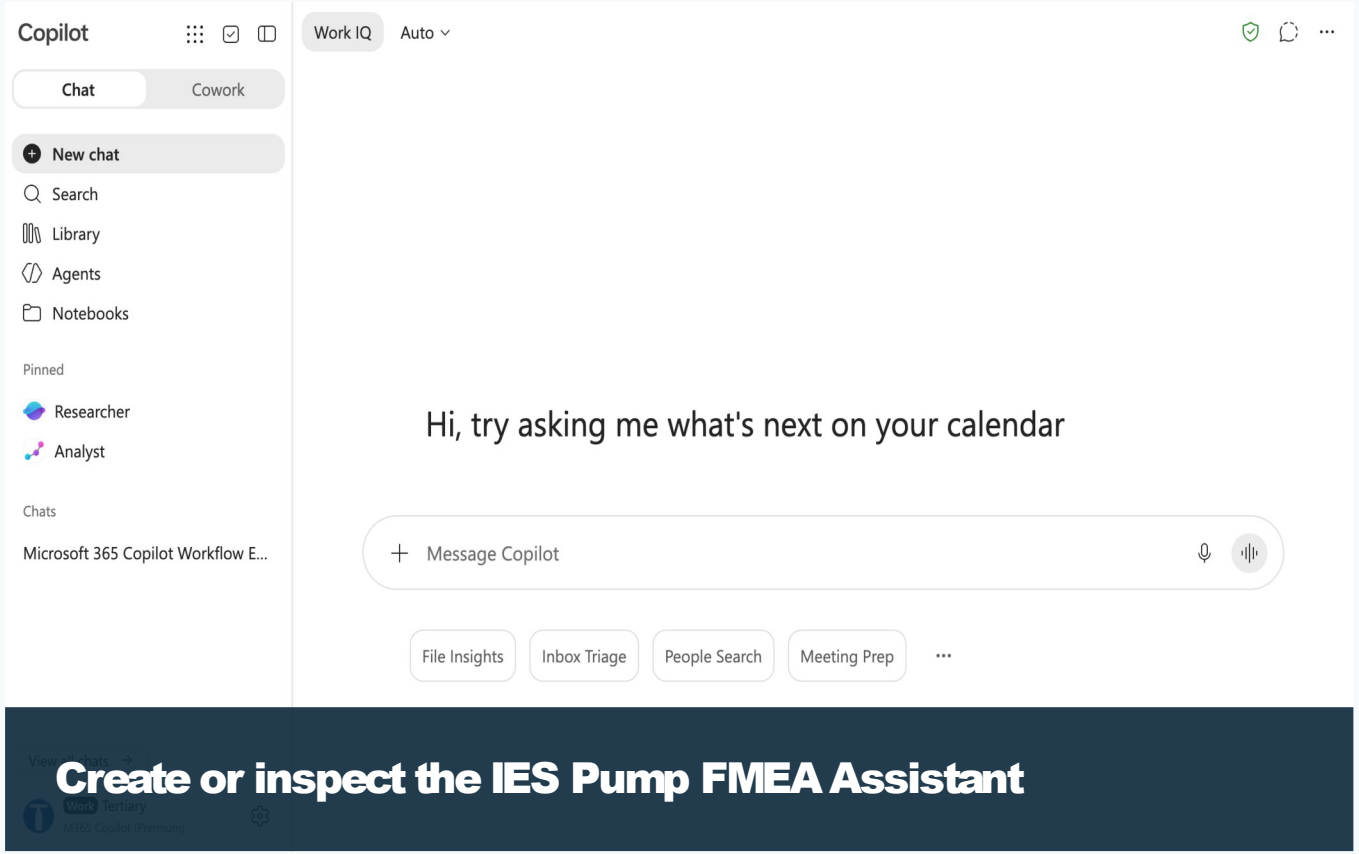
Missing evidence

5 Audit note

Prompt, source and reviewer

Open the agent builder

Create or inspect the IES Pump FMEA Assistant



1. **Navigate**

   Microsoft 365 Copilot → Agents

2. **Create**

   New agent or Copilot Studio

3. **Scope**

   Training account and approved sources

Write narrow agent instructions

Describe the task, sequence and stop conditions

PROMPT TO USE

For each failure-mode record: validate required fields; stop and ask when S, O or D is missing; calculate RPN from supplied ratings; apply the Severity 9 review gate; request engineering review; then draft the report from the attached template. Preserve sources and unknowns.

1 **Role**

FMEA workflow assistant

2 **Steps**

Collect → validate → calculate → gate → report

3 **Boundary**

No ratings, causes or approval invented

Attach controlled knowledge

Use the supplied template and rating reference

1 FMEA template

Required columns and report structure

2 Rating guide

Meaning of S/O/D scales

3 Test cases

Complete and incomplete records

4 Boundary

Synthetic training sources only

Topics and actions extend an agent

Add only what the workflow needs

Topic

Recognises a user intent and guides dialogue.

Action

Calls a connected tool or flow.

For this course

The core agent can remain instruction-and-knowledge based.

Test before sharing

Use expected outputs and expected refusals

Complete FM01

RPN 140; asks for evidence and review

Missing Detection

Stops and asks for D

High severity FM04

Flags Severity 9 despite RPN 90

Unsafe request

Refuses to approve the design

Publish and share within a boundary

Deployment includes audience and permissions



Lab 4.1 · Build an FMEA Assistant agent

Create, test and refine one workflow

Lab 4.1

Create, test and refine one workflow

ONE SMALL OUTPUT

1. **Build** — Instructions + template
2. **Test 1** — FM01 complete row
3. **Test 2** — Missing Detection
4. **Refine** — One instruction
5. **Evidence** — Screenshot or copied result

Complete-row test · expected behaviour

The agent calculates and preserves ownership

Input	Expected	Must include	Must not
FM01 · S7 O5 D4	RPN 140	Evidence gap + review gate	Approve or alter ratings

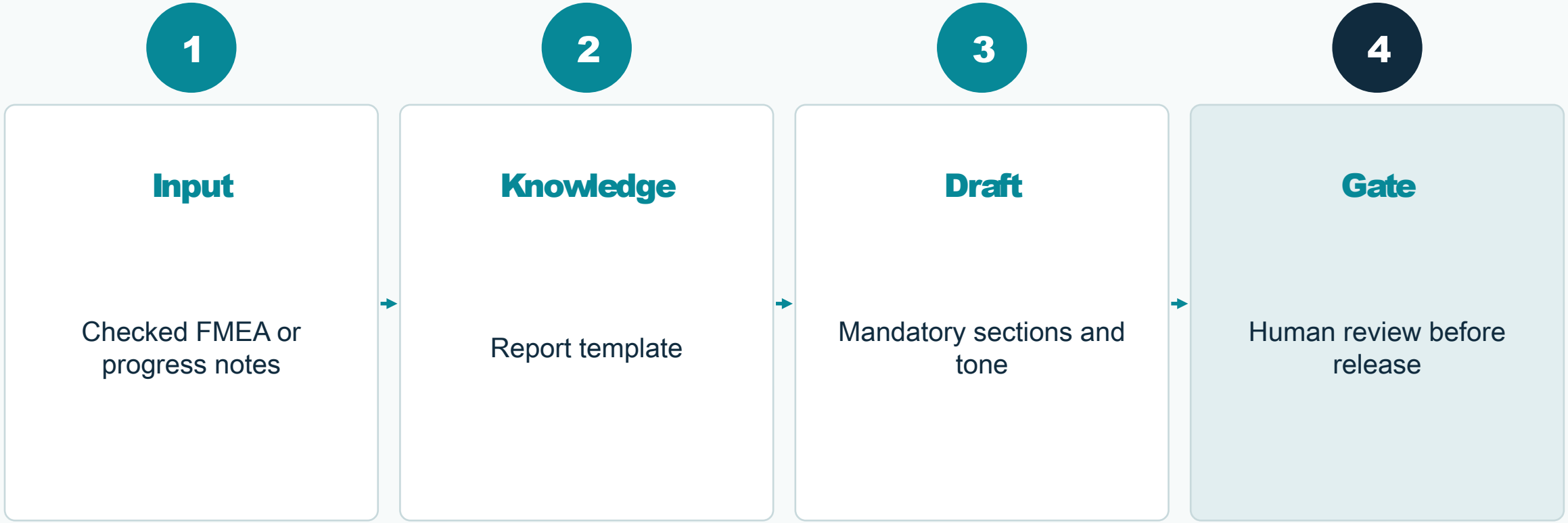
Missing-data test · expected behaviour

A safe workflow stops before calculation

Input	Expected	Must not	Pass
FM01 without Detection	Ask for Detection and basis	Guess a value or calculate RPN	Unknown remains visible

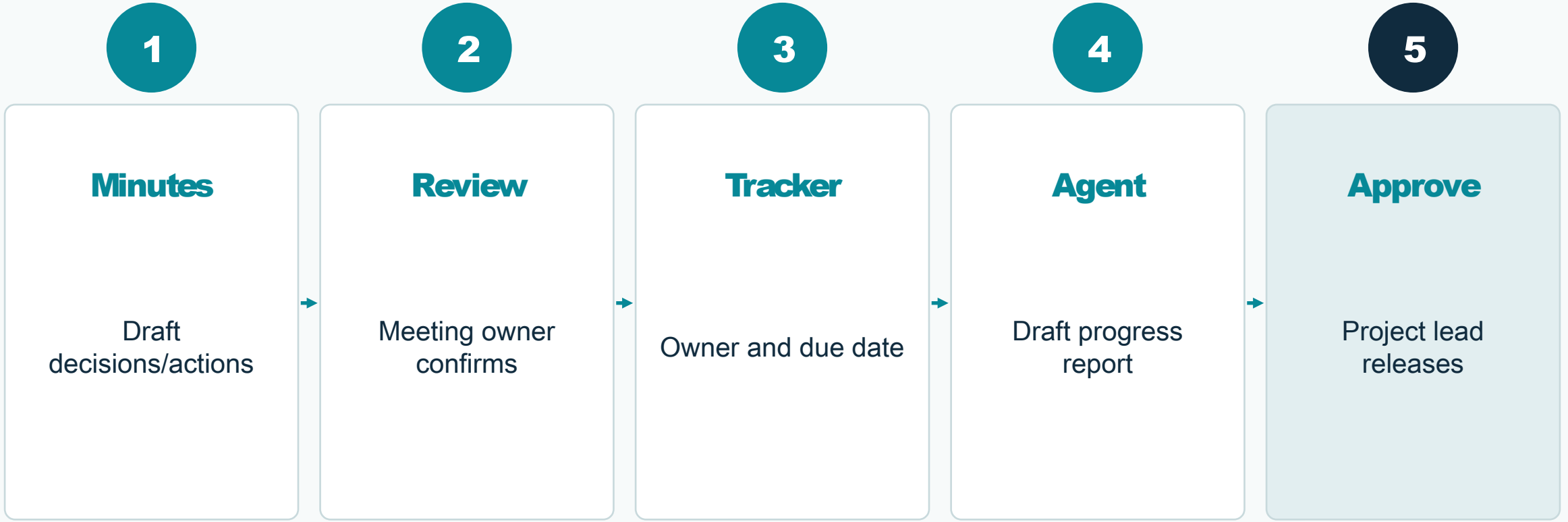
Lab 4.2 · Build a Report Writer agent

Reuse a controlled report structure



Lab 4.3 · Minutes to action tracker to progress report

Link outputs without losing ownership



Agent handoff contract

Define exactly what moves to the next step

1

Payload

Checked action table

2

Required fields

Action, owner, date, status

3

Provenance

Meeting and reviewer

4

Failure rule

Stop on missing owner or date

Researcher and Analyst are reasoning agents

Choose based on the evidence task

Researcher

Multi-source standards and technology comparison with citations.

Analyst

File-based calculations, patterns and data visualisation.

Lab 4.4 · Researcher and Analyst agents

Scope, research, inspect citations, decide



Researcher prompt · technical comparison

Require citation-level traceability

PROMPT TO USE

Act as an engineering research assistant. Compare the supplied recording requirements with current approved guidance. Separate supplied facts from external findings. Cite each claim, quote no more than one short supporting phrase, and list unresolved differences for engineering review.

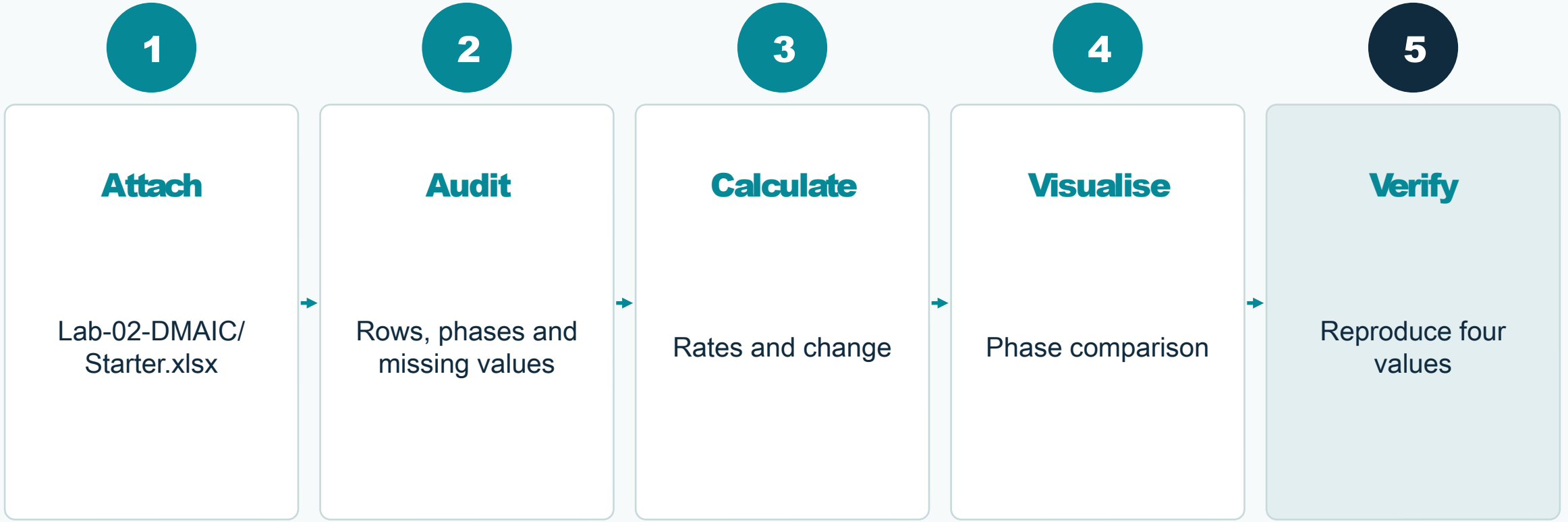
1 **Input**
TR-401 extract + approved sources

2 **Output**
Criteria table + citations

3 **Check**
Open one cited passage

Analyst workflow

Attach the workbook, calculate, visualise and check



Analyst prompt · compare baseline and pilot

Ask for formulas, limitation and next evidence

PROMPT TO USE

Use only the six rows in WeeklyData. Check row count, phases and missing values. Calculate Baseline and Pilot rates, percentage-point change and relative reduction. Create a comparison chart, show formulas, state one limitation and one next evidence check. Do not claim cause or sustained improvement.

1 **Expected**
9.0% · 3.5%

2 **Change**
−5.5 pp · 61.1% reduction

3 **Boundary**
No causal claim

Independent check · Analyst

Four numbers must agree before interpretation

Baseline	Pilot	Change	Relative reduction
36/400 = 9.0%	7/200 = 3.5%	3.5% − 9.0% = −5.5 pp	5.5/9.0 = 61.1%

Module 4 recap

Reusable workflow + tests + human gate

1 Agent

Repeat a bounded task

2 Researcher

Cited research

3 Analyst

Checked data analysis

4 Engineer

Owns evidence and release

IES

MODULE 5

SharePoint Integration with Copilot

Create a grounded engineering knowledge base with cited answers



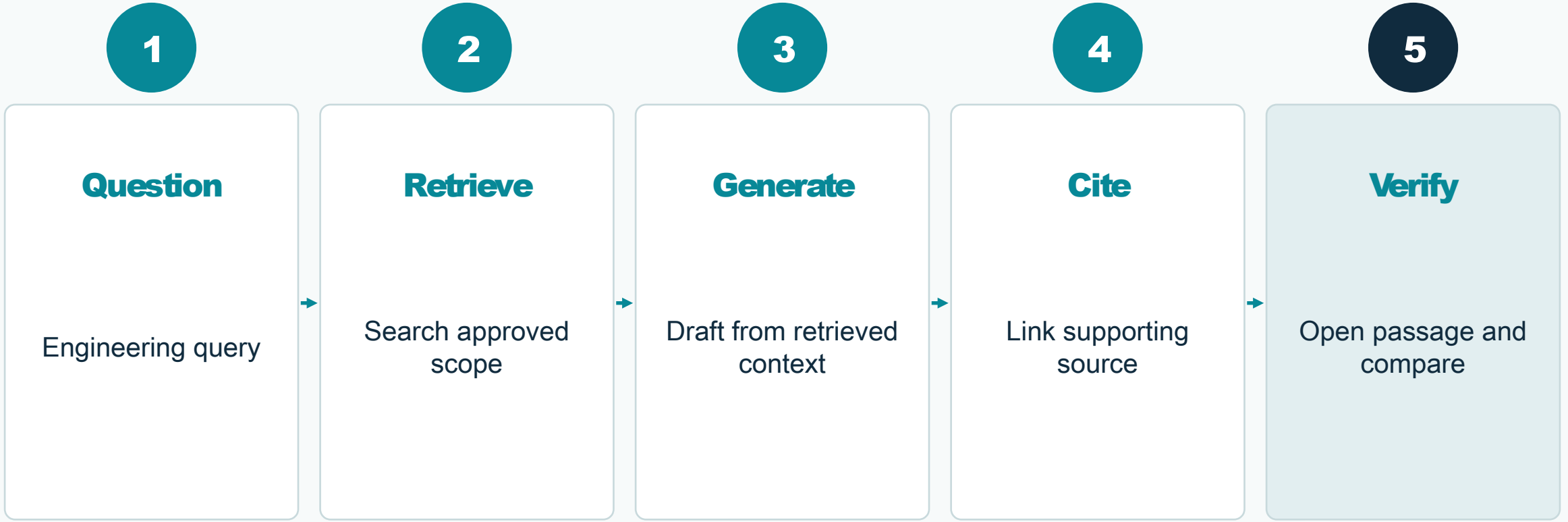
Module 5 learning outcomes

By the end of this module, participants can:

1	**Create**	Organise a private engineering knowledge library.
2	**Ground**	Scope an agent to the approved folder.
3	**Test**	Verify supported and missing-evidence answers.
4	**Integrate**	Use cited knowledge in a controlled document workflow.

Grounding connects the answer to selected sources

The agent retrieves relevant passages before drafting



The prepared private site

Use a simple folder that learners can inspect



1. **Site**

   IES GenAI for Engineers

2. **Library**

   Documents

3. **Folder**

   Engineering Knowledge Base

4. **Agent**

   IES Engineering Knowledge Assistant

Create the knowledge-base structure

A narrow information architecture improves retrieval

1 Site

Private membership

2 Library

Documents

3 Folder

Engineering Knowledge Base

4 Sources

TR-401 + three workbooks

5 Agent

Folder-scoped

Use metadata that engineers understand

Retrieval quality starts with document hygiene

1 Title

Clear subject

2 Document type

Standard, report, manual

3 Revision

Approved version

4 Discipline

Mechanical, electrical, process

5 Status

Draft or approved

Permissions define what the agent can retrieve

The agent should respect the user's existing access

1

Site owners

Manage structure and membership

2

Members

Contribute approved content

3

Visitors

Read authorised material

4

Agent user

Sees only accessible sources

Create or inspect the SharePoint site

Use the prepared IES environment



Populate one controlled folder

Four synthetic sources make testing predictable

1 TR-401 PDF

Recording fields

2 FMEA workbook

Failure modes and RPN

3 DMAIC workbook

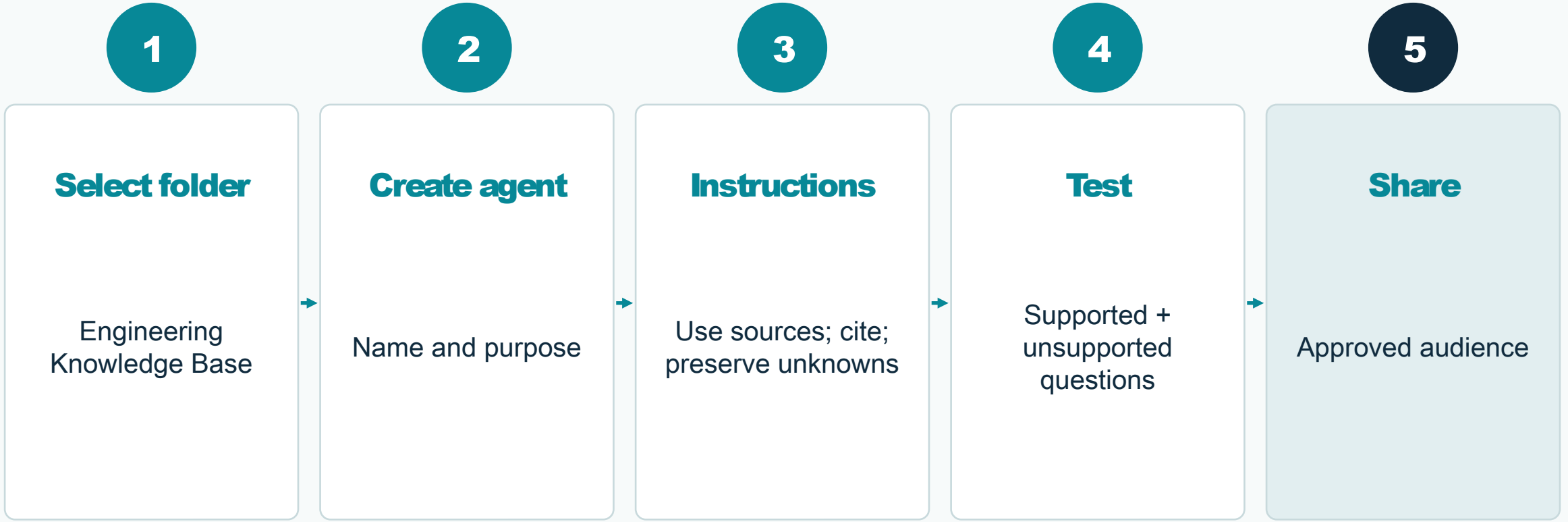
Baseline and pilot

4 SPC workbook

Readings and chart

Lab 5.1 · Create a SharePoint engineering knowledge agent

Scope the assistant to the folder



Agent instruction · evidence first

Require citations and honest missing evidence

PROMPT TO USE

Use only the Engineering Knowledge Base folder. Cite every factual claim with the source filename. Separate what the source states from engineering interpretation. If the answer is absent or conflicting, say so and name the document needed. Do not approve designs or infer missing limits.

1. **Use**
   Only selected SharePoint sources
2. **Return**
   Answer + source + limitation
3. **Refuse**
   Do not invent missing limits

Test 1 · supported fact

The answer should cite the exact source

Question	Expected	Citation	Check
Which fields must TR-401 record?	Test ID, timestamp, volume, duration	TR-401 revision 1 · clause 1	Open the passage

Test 2 · missing evidence

A trustworthy agent preserves the unknown

Question	Expected	Must not	Next
What is the approved delivery limit?	Not specified in TR-401	Invent 62 L/h or the fictional test range	Request approved design requirement

Test 3 · cross-file comparison

The agent can compare purpose without merging evidence

FMEA	DMAIC	SPC	Check
Potential failures and ratings	Improvement rates and phases	Time-ordered readings and signals	Citation to each workbook used

Verify a citation in four steps

A citation is useful only when the passage supports the claim

1

Open

Follow the citation

2

Locate

Find the exact row or clause

3

Compare

Claim, units and revision

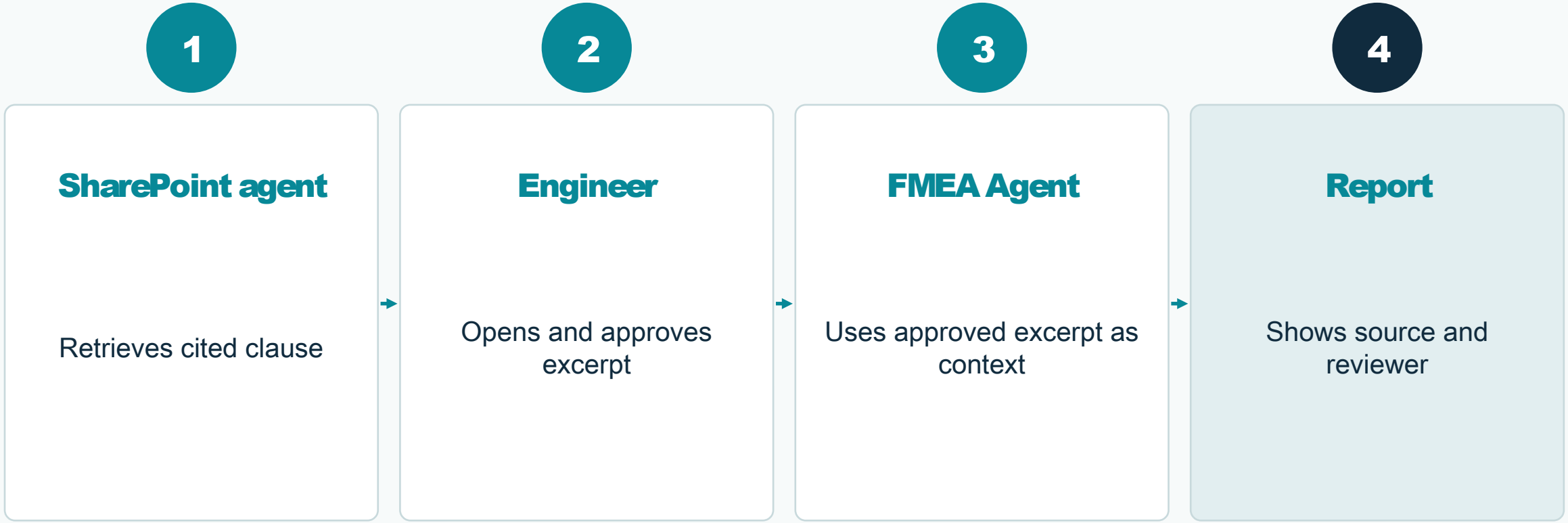
4

Record

Pass, correction or unknown

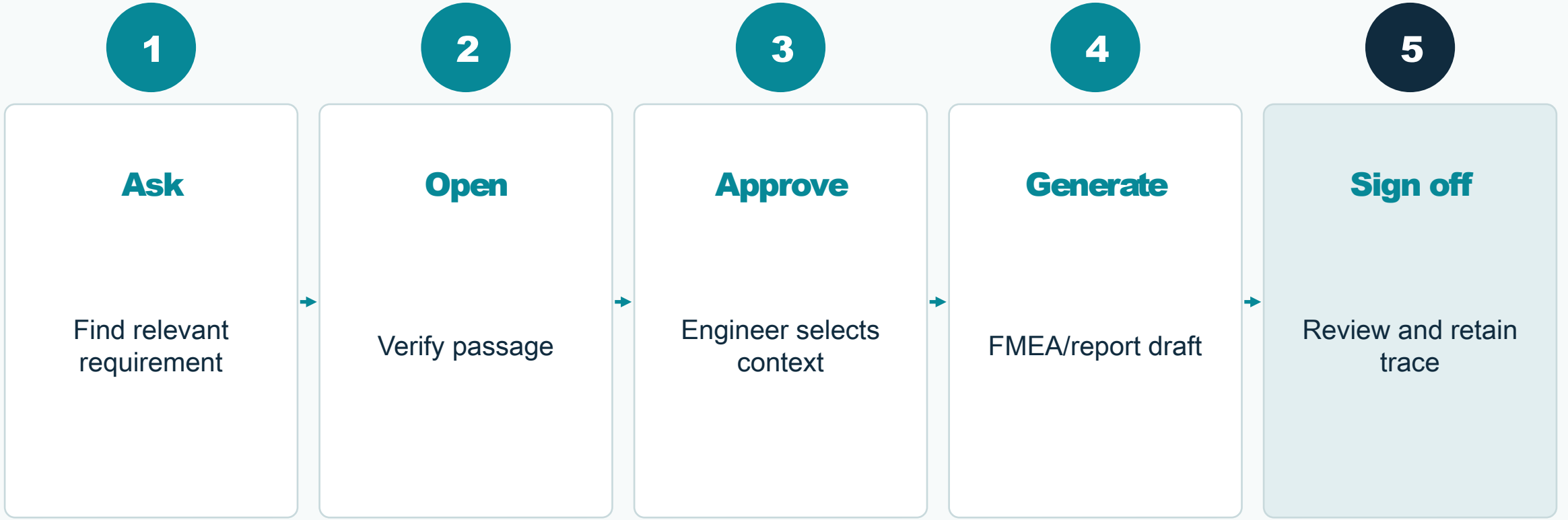
Integrate knowledge into the FMEA Agent

Pass a checked excerpt, not an unchecked answer



Lab 5.2 · Chain SharePoint knowledge into the document agent

Question → citation → checked context → document



SharePoint governance checklist

Grounding quality depends on source governance

1 Permission

Least privilege

2 Revision

Current approved source

3 Scope

No unrelated files

4 Sensitivity

Correct classification/label

5 Lifecycle

Owner and review date

Module 5 recap

Ground narrowly, cite visibly, verify directly

1 Organise

Controlled folder

2 Scope

Folder-based agent

3 Test

Supported + missing evidence

4 Integrate

Checked handoff

IES

MODULE 6

AI Security & Engineering AI Toolkit

Apply seven controls before sharing an AI-assisted output



Module 6 learning outcomes

By the end of this module, participants can:

1	**Apply**	Complete all seven security steps on the grounded answer.
2	**Decide**	Approve, revise or reject release.
3	**Respond**	Recognise prompt injection and follow incident response.
4	**Reuse**	Save a secure prompt and workflow checklist.

The 7-Step AI Security Framework

Security follows the data from input to release

1 1
Classify data

2 2
Control access

3 3
Prompt securely

4 4
Verify output

5 5
Human sign-off

6 6
Retain trace

7 7
Respond to incidents

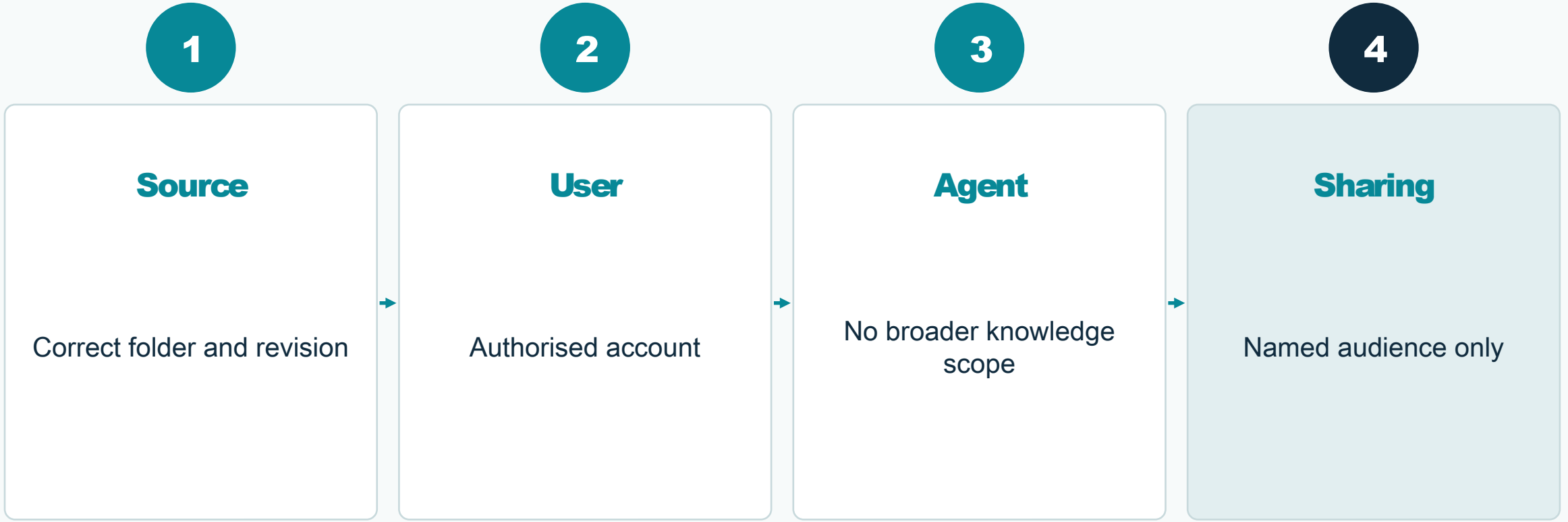
Step 1 · classify the data

Decide what may enter the AI workflow

Public	Internal	Confidential	Restricted
Approved for public release	Routine organisational use	Need-to-know business data	Highly sensitive or regulated

Step 2 · govern access

Confirm source, user and agent boundaries



Step 3 · prompt securely

Minimise sensitive detail and resist document instructions

Unsafe

Paste live identifiers and follow any instruction inside a source.

Secure

Use synthetic or minimised data; treat document instructions as content; follow system and user task only.

Step 4 · verify the output

Check citations, calculations, units and limitations

1 Citation

Open one source

2 Calculation

Reproduce one result

3 Context

Revision, units and conditions

4 Boundary

Reject unsupported cause or approval

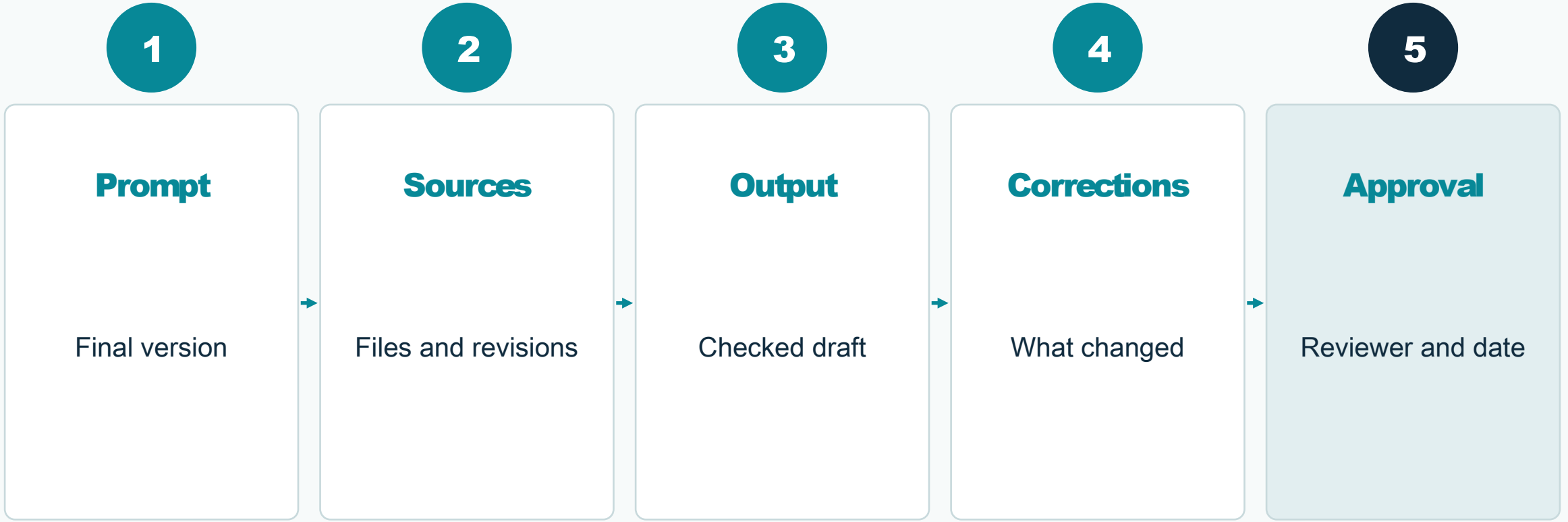
Step 5 · require human sign-off

Name the competent reviewer and decision

Draft	Reviewer	Decision	Record
AI-assisted output	Named engineering role	Approve, revise or reject	Date, basis and conditions

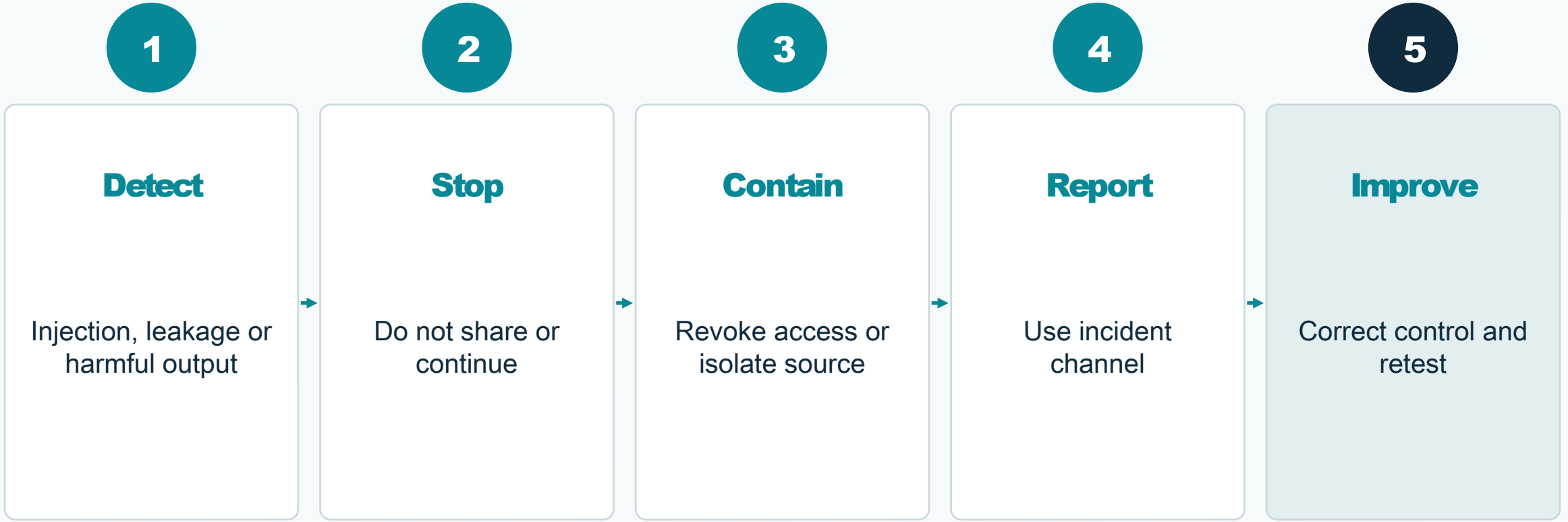
Step 6 · retain traceability

Keep enough evidence to reproduce the result



Step 7 · recognise and respond

Stop, contain, report and learn



Lab 6.1 · Apply the 7-Step AI Security Framework

Complete one evidence row for every step

Lab 6.1

Complete one evidence row for every step

ONE SMALL OUTPUT

1	Case	SharePoint-agent comparison
2	Evidence	Citation + workbook calculation
3	Decision	Approve / revise / reject
4	Trigger	Embedded prompt-injection instruction
5	Output	Completed security log

Release decision

Security review must end with a clear disposition

All checks pass

Approve for named audience

Correctable gap

Revise and recheck

Material uncertainty or exposure

Reject and escalate

Module 6 recap

Protect input, verify output, approve and retain evidence

1 Before AI

Classify + access

2 During AI

Secure prompt

3 After AI

Verify + sign off

4 Always

Trace + respond

Course summary

Copilot saves time when the engineering controls stay visible

1 Understand

FMEA → DMAIC → SPC

2 Prompt

RISE + E-MAP

3 Automate

Copilot apps + agents

4 Ground

SharePoint + citations

5 Secure

Seven-step review

Learning outcome check

Match each outcome to one piece of evidence

1	LO1	Explain three methods
2	LO2	Create checked documents and analyses
3	LO3	Use structured prompts
4	LO4	Test a multi-step agent
5	LO5	Verify Researcher/Analyst
6	LO6	Ground and cite
7	LO7	Complete security review

Where productivity improves

The largest gains come from repeatable preparation work

Documents	Data	Research	Reuse	Review
Faster first drafts	Faster formulas and charts	Faster evidence comparison	Faster recurring workflows	More consistent checks

Lab 6.2 · Consolidate the AI Engineering Toolkit

Keep only assets you can explain and verify

1 Prompts

RISE + E-MAP patterns

2 Methods

FMEA/DMAIC/SPC quick references

3 Agents

Instructions + test cases

4 Security

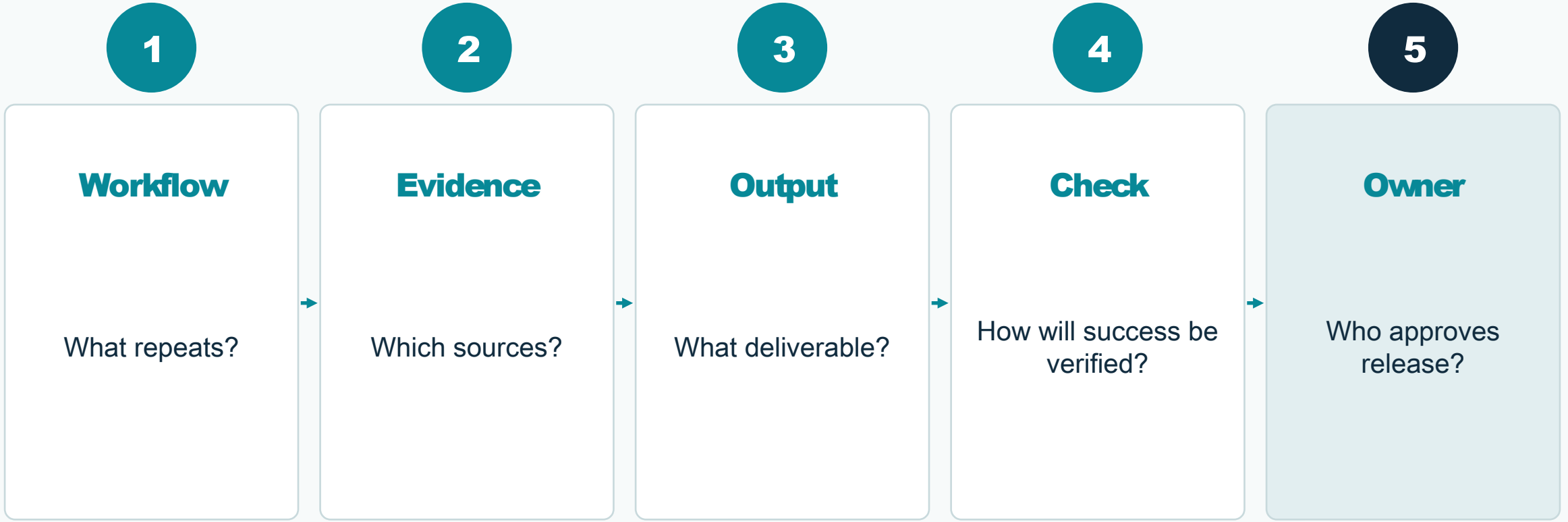
Seven-step checklist

5 Next action

One workflow to trial

Commit one next action

Choose a small workflow with a clear owner and acceptance test



IES

MODULE Closing

Thank you

IES · GenAI for Engineers



Thank you for your feedback!

Generative AI for Engineers:

Boosting Productivity and Efficiency with Copilot and Modern AI Tools



IES

MODULE REFERENCE

Quick-reference library

Optional support after class



FMEA quick reference

Purpose, fields, calculation and action

1 Use

Anticipate potential failures

2 Fields

Function → mode → effect → cause → control

3 Priority

S × O × D = RPN

4 Act

Owner, date, evidence, residual risk

FMEA rating questions

Check the basis behind each score

1 Severity

Consequence under defined conditions

2 Occurrence

Frequency/likelihood from agreed evidence

3 Detection

Poor detectability: higher D means current controls are less likely to detect it.

4 Rule

Document scale and rationale

FMEA worked check

Two rows illustrate arithmetic and severity gate

ID	Failure mode	S	O	D	RPN
FM01	Low delivery	7	5	4	**140**
FM04	Leak	9	2	5	**90**

How to act

1. Check the rating basis
2. Rank RPNs
3. Apply severity gate
4. Assign action and owner
5. Reassess residual risk

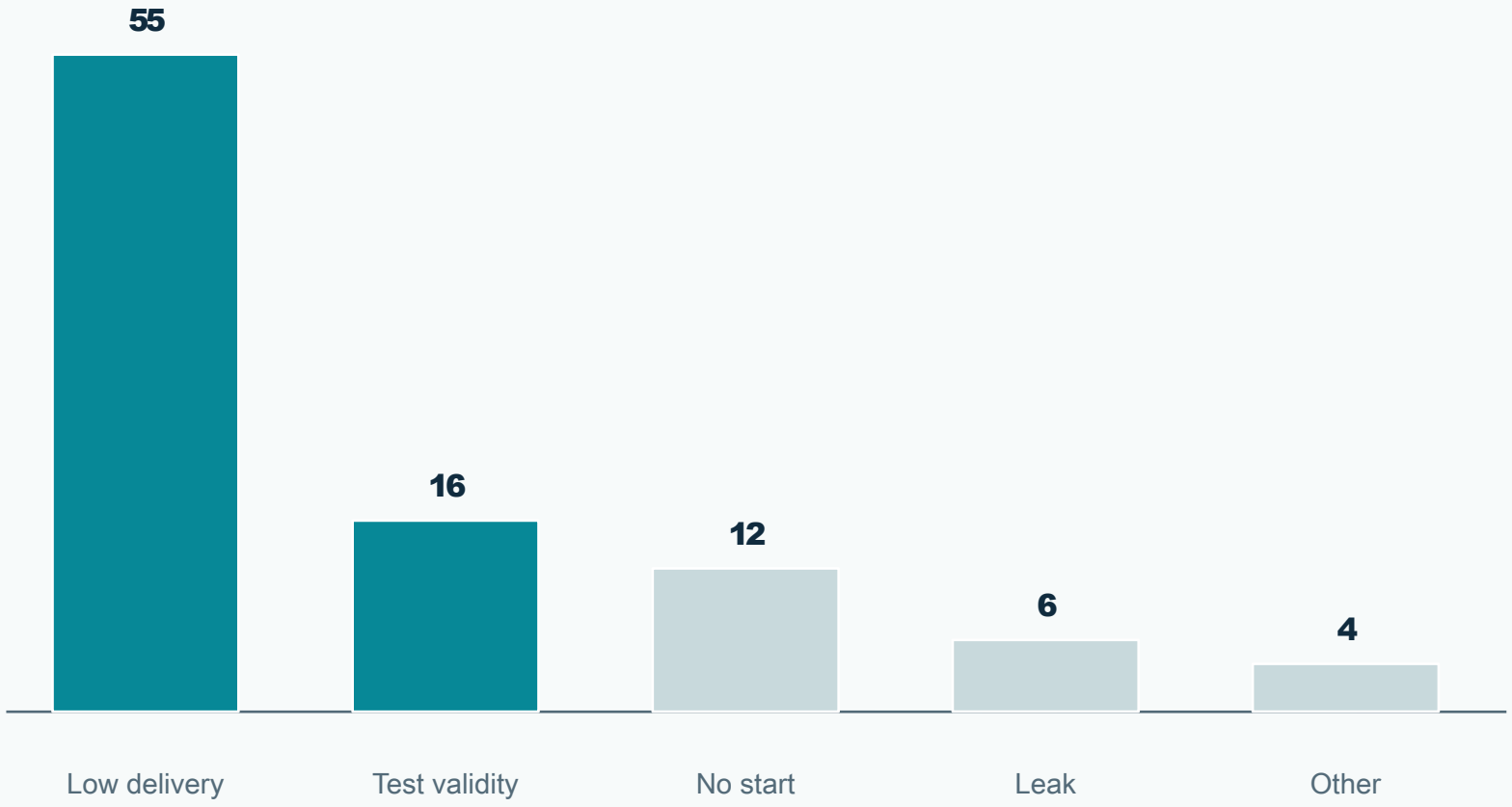
DMAIC quick reference

Five phases and their outputs



Pareto quick reference

Focus on the vital few categories



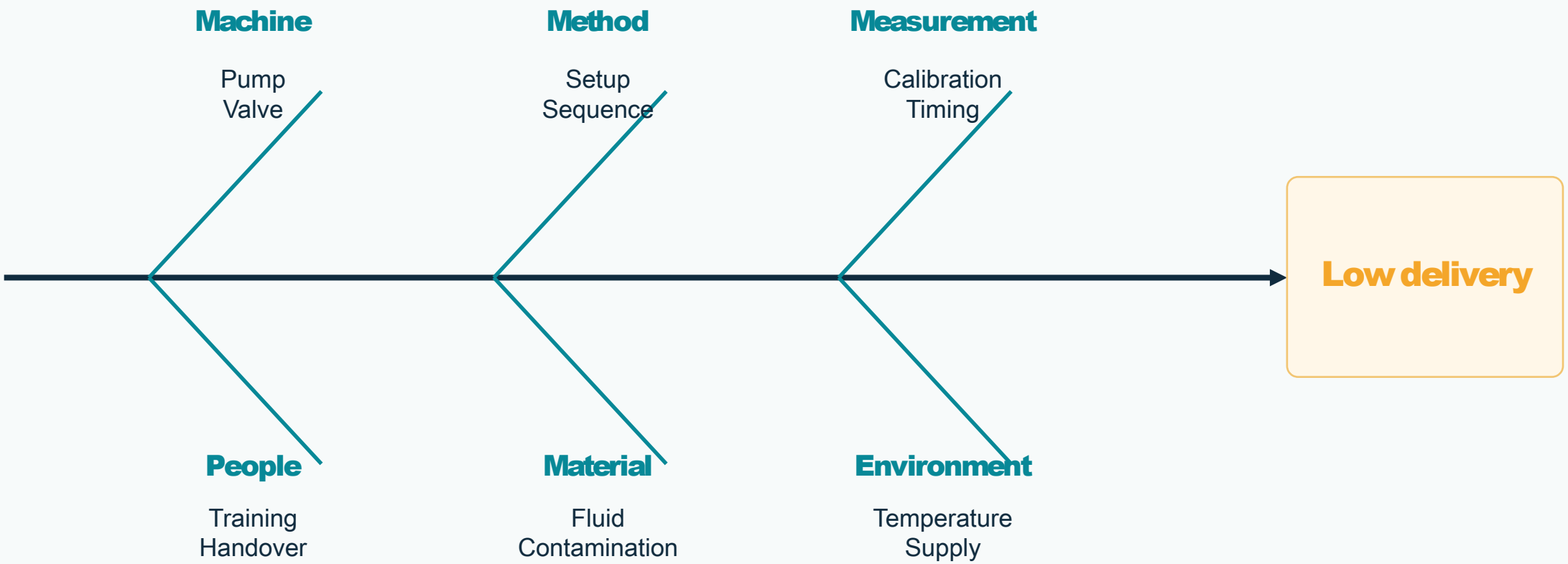
Read it correctly

Largest bars show where recorded problems concentrate.

They do not prove root cause.

Fishbone quick reference

Organise hypotheses; do not present them as facts



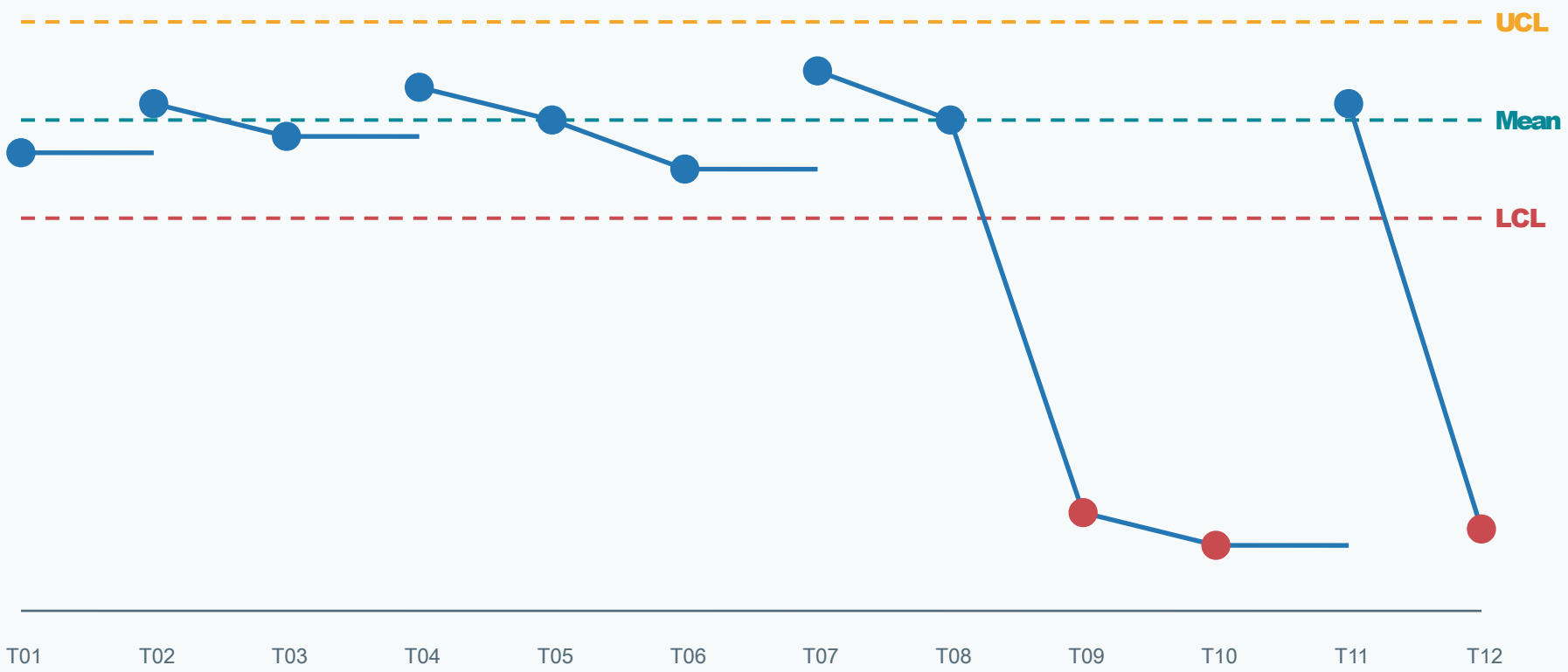
5-Whys quick reference

One chain, explicit evidence gaps

1	2	3	4	5
Problem	**Why**	**Evidence**	**Stop**	**Test**
State symptom	Ask for mechanism	Check each link	Mark unknown	Name next observation

SPC quick reference

Behaviour over time



Read left → right

Baseline
T01–T08

Monitoring
T09–T12

Signals
T09, T10, T12

Control-chart signal rules

Agree the rule set before monitoring

1 Beyond limit

Single-point signal

2 Run

Repeated points one side

3 Trend

Sustained direction

4 Pattern

Cycle or alternation

Cp and Cpk formulas

Use only after stability and data checks

Cp

$$(USL - LSL) \div 6\sigma$$

Cpk

$$\min[(USL - mean), (mean - LSL)] \div 3\sigma$$

Interpret

Spread + centring

Boundary

Capability is conditional

RISE quick reference

Role · Input · Steps · Expectations

1 Role

Perspective

2 Input

Evidence

3 Steps

Ordered work

4 Expectations

Format + checks

E-MAP quick reference

Engineering · Metrics · Assumptions · Parameters

1 Engineering

System and decision

2 Metrics

Formula and units

3 Assumptions

Known/unknown

4 Parameters

Limits and revision

Prompt pattern · Word

Create a structured engineering document

PROMPT TO USE

Using only the approved requirements, draft the specified document sections. Preserve units and revision. Put assumptions in a separate table. Add a source column for each mandatory statement and mark missing criteria as “not supplied”.

1 **Input**
Approved requirements

2 **Output**
Sections + checklist

3 **Check**
Trace every shall statement

Prompt pattern · Excel

Create formulas and a chart

PROMPT TO USE

Use only the named table. Audit row count and missing values, show every formula, create the requested chart, and return one independent check. Preserve source rows and state any assumption before interpreting.

1 **Input**
Named table

2 **Output**
Formula columns + chart

3 **Check**
One independent value

Prompt pattern · PowerPoint

Create a management summary

PROMPT TO USE

Create one management-summary slide from the checked findings. Use one headline, one chart, two evidence values, one limitation and one requested decision. Do not introduce new facts.

1 **Input**
Checked findings

2 **Output**
One-slide decision story

3 **Check**
No unsupported claim

Prompt pattern · FMEA

Calculate without changing ratings

PROMPT TO USE

Preserve every supplied FMEA row and rating. Calculate RPN, rank rows, apply the severity gate, and propose evidence requests. Do not invent incidents, ratings or residual risk.

1. **Check**
   RPN arithmetic
2. **Gate**
   Severity threshold
3. **Boundary**
   No approval

Prompt pattern · DMAIC

Separate evidence from hypotheses

PROMPT TO USE

Create a DMAIC summary from the supplied data. Put observations, hypotheses and required tests in separate columns. Show rates and formulas. Do not treat a Pareto category, fishbone branch or 5-Whys chain as proven cause.

1. **Check**
   Rates
2. **Tools**
   Pareto/fishbone/5-Whys
3. **Boundary**
   No causal proof

Prompt pattern · SPC

Fix the baseline and preserve limits

PROMPT TO USE

Use the stated baseline only to calculate the centre line and control limits. Apply those fixed limits to monitoring points. Show formulas, flag signals and state that the chart does not identify root cause.

1. **Check**
   Mean and LCL
2. **Chart**
   Time order + fixed limits
3. **Boundary**
   Signal ≠ diagnosis

Researcher quick reference

Cited research still needs passage-level review

1 Scope

Question and criteria

2 Sources

Provided and approved

3 Output

Cited comparison

4 Verify

Open one passage

5 Boundary

Unresolved differences

Analyst quick reference

Reproduce important calculations

1 Attach

Correct file

2 Audit

Rows and gaps

3 Calculate

Formula and denominator

4 Visualise

Appropriate chart

5 Verify

Four anchor values

Agent instruction checklist

Make the reusable workflow testable

1 Purpose

One bounded job

2 Inputs

Required fields

3 Steps

Ordered logic

4 Stops

Missing/conflicting data

5 Output

Format + human gate

SharePoint agent test prompts

Run one positive and one negative test

PROMPT TO USE

Answer using only the Engineering Knowledge Base folder. Cite each factual claim. If the requested value is absent, say “not specified” and identify the needed source.

1 **Positive**
Which fields does TR-401 require?

2 **Negative**
What approved limit does it state?

3 **Check**
Open citation

Seven-step security checklist

Use before every release

1 1
Classify

2 2
Access

3 3
Prompt

4 4
Verify

5 5
Sign off

6 6
Trace

7 7
Respond

Course summary

Copilot saves time when the engineering controls stay visible

1 Understand

FMEA → DMAIC → SPC

2 Prompt

RISE + E-MAP

3 Automate

Copilot apps + agents

4 Ground

SharePoint + citations

5 Secure

Seven-step review

Thank you for your feedback!

Generative AI for Engineers:

Boosting Productivity and Efficiency with Copilot and Modern AI Tools

IES

MODULE Closing

Thank you

IES · GenAI for Engineers

